

How Excess Latent Dimensionality Delays Memorization in Diffusion Models

Anonymous Authors¹

Abstract

We study how the gap between latent and intrinsic dimensionality shapes memorization in diffusion models. Extending recent spectral analysis to anisotropic latent data, we find that excess null dimensions populate a new “noise-dimension” bulk in the score network’s feature-correlation spectrum, sitting between population-driven and sample-specific eigenmodes. Because gradient flow learns modes in decreasing-eigenvalue order, this bulk acts as a buffer: each excess latent direction adds a mode that must be absorbed before sample-specific memorization begins. Synthetic random-feature and MLP experiments isolate this mechanism. On CelebA and CIFAR-10 latent diffusion, memorization is non-monotone in d_{latent} : it first rises as the VAE becomes expressive enough to reconstruct the training subset, then falls at larger latent widths while FID is best at intermediate dimensions. Finally, a data-spectrum surrogate computed from the frozen VAE encoder predicts much of the empirical memorization trajectory, suggesting a route from latent geometry to stopping-time estimates.

1. Introduction

Diffusion models are the dominant generative model class for high-dimensional continuous data such as images, video, audio, and molecules (Ho et al., 2020; Song et al., 2021; Rombach et al., 2022; Karras et al., 2022; Lee et al., 2025). Sampling corresponds to numerically integrating the time-reversed stochastic differential equation associated with a forward noising process; the only learnable component is a score network $s_\theta(x_t, t) \approx \nabla \log p_t(x_t)$ trained to match the noisy marginals of the data distribution. Sampling can equivalently be expressed as a deterministic ordinary differential equation with the score as a velocity field, an approach

taken by flow-matching methods (Lipman et al., 2023).

Latent diffusion models (Rombach et al., 2022; Esser et al., 2024; Podell et al., 2024) run forward and backward diffusion not in pixel space but in a learned latent space of much smaller dimensionality d_{latent} , typically obtained from a VAE encoder. By placing diffusion in a representation closer to the data’s true intrinsic dimensionality $d_{\text{intrinsic}}$ (Pope et al., 2021; Brown et al., 2023; Stanczuk et al., 2024; Kamkari et al., 2024), though still well above it (with $d_{\text{latent}} \gg d_{\text{intrinsic}}$ in practice), latent diffusion achieves comparable or better sample quality at a fraction of the compute of pixel-space diffusion.

A model achieving zero training loss without regularization learns the empirical score and reproduces training samples at the end of the backward process (Gu et al., 2025); the regime kicks in when the training set is small relative to model capacity and disappears past a dataset-size threshold (Kadkhodaie et al., 2024). What is missing is a sharp account of how this transition depends on the latent-space dimensionality, the design knob that distinguishes latent-diffusion from pixel-space diffusion. We ask: *how does the gap between d_{latent} and $d_{\text{intrinsic}}$ shape the transition from generalization to memorization, and is there a mechanism a practitioner can tune by adjusting the latent width?*

Existing theory provides a starting point. Bonnaire et al. (2025) characterize diffusion training as a competition between a generalization time τ_{gen} , at which the score network learns the population distribution, and a memorization time τ_{mem} , at which it begins fitting individual training points. Their analysis assumes isotropic data covariance, which collapses the distinction between the $d_{\text{intrinsic}}$ -dimensional signal subspace and the $d_{\text{latent}} - d_{\text{intrinsic}}$ null directions of the latent space. We extend their spectral analysis to the latent-diffusion regime $d_{\text{latent}} > d_{\text{intrinsic}}$, identify a new noise-dimension bulk that opens between the population and sample bulks, and show that it acts as a buffer that delays memorization.

Contributions.

- We extend Bonnaire’s two-bulk theorem to anisotropic data and identify a four-bulk eigenvalue structure for the score network’s feature correlation matrix, with bulk-count boundaries that land exactly at indices

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

$d_{\text{intrinsic}}$ and d_{latent} .

- We identify the noise-dimension bulk as a memorization buffer: increasing d_{latent} adds $d_{\text{latent}} - d_{\text{intrinsic}}$ extra modes that the readout must absorb before reaching the sample bulk, delaying τ_{mem} while leaving τ_{gen} approximately fixed.
- We confirm the buffer mechanism in trainable MLP score networks on synthetic data with five-seed confidence intervals, holding hidden width fixed so the d_{latent} sweep is not a model capacity sweep.
- We test the same idea in VAE latent spaces on CelebA and CIFAR-10. Memorization first rises as the VAE becomes faithful, then decreases at high d_{latent} , while FID is best at intermediate latent widths.
- We introduce a data-only spectral predictor: compute the VAE latent covariance spectrum, calibrate a single global pressure curve, and overlay predicted memorization thresholds on empirical memorization trajectories.

2. Setup and theory roadmap

Question. We study diffusion in a latent space of dimension d_{latent} when the data occupy an effective $d_{\text{intrinsic}}$ -dimensional subset. The central object is the gap between the time at which a score model learns population structure, τ_{gen} , and the time at which it begins to fit individual training samples, τ_{mem} . The working hypothesis is that the excess width $d_{\text{latent}} - d_{\text{intrinsic}}$ is not inert: it creates a sequence of low-variance directions that the model must traverse before reaching sample-specific modes.

Synthetic data choices. We use Gaussian-mixture data because it gives us the cleanest control over the mechanism. A k -component mixture is sampled in a signal subspace of dimension $d_{\text{intrinsic}}$, embedded into an ambient latent space of dimension d_{latent} , and padded with isotropic null-direction variance $\sigma_{\perp}^2$. The signal variance and cluster scale control the population bulk; the null variance controls the buffer bulk. This construction gives an analytic score for the trainable MLP, exact knowledge of which coordinates are signal versus null, and controlled sweeps in either d_{latent} at fixed $d_{\text{intrinsic}}$ or $d_{\text{intrinsic}}$ at fixed d_{latent} .

RFNN theory. For the random-feature network, freezing the first layer makes the score-matching loss quadratic in the readout. Gradient flow then diagonalizes in the eigenbasis of the empirical feature covariance U , and each mode is learned on timescale $1/\lambda_i$. The theory section below derives the qualitative prediction we need: anisotropic data split the usual population/sample spectrum into a signal bulk, a noise-dimension buffer bulk, a sample bulk, and a rank-null tail. A closed-form theorem for the bulk edges is still a TODO; the current paper uses the RFNN eigenspectra as the mechanistic microscope.

Trainable MLP and real latents. We then ask whether the same picture survives after replacing the frozen RFNN with a trainable MLP score model. In synthetic data, the hidden width is held fixed across the d_{latent} sweep so parameter count and per-step FLOPs stay roughly constant. In real data, we train VAEs on CelebA and CIFAR-10, encode a diverse 1k-image subset, and train the same latent-space MLP diffusion model for 5M steps over five seeds. We measure memorization with the [Somepalli et al. \(2023\)](#) nearest-neighbor ratio test and quality with FID.

3. Synthetic data: RFNN spectral microscope

3.1. Setup and assumptions

We adopt the RFNN framework of [Bonnaire et al. \(2025\)](#) unchanged: a two-layer network

$$s_A(x_t) = \frac{1}{\sqrt{p}} A \tanh\left(\frac{1}{\sqrt{d_{\text{latent}}}} W x_t\right), \quad (1)$$

with frozen first-layer weights $W \in \mathbb{R}^{p \times d_{\text{latent}}}$ drawn i.i.d. from $\mathcal{N}(0, 1)$, $\tanh$ activation, and a trained readout $A \in \mathbb{R}^{d_{\text{latent}} \times p}$ initialized to zero. Training is score matching at a single fixed diffusion time $t = 0.01$, and the analysis is taken in the proportional asymptotic limit $d_{\text{latent}}, n, p \rightarrow \infty$ with the relevant aspect ratios held fixed. In the clean figures we report both a controlled-width choice, $p = d_{\text{latent}} + n + 300$, which keeps the rank-null tail visible, and fixed-ratio/fixed-width controls in the appendix.

Training dynamics. Freezing W makes the score-matching loss quadratic in A , so full-batch gradient flow on A is the linear ODE

$$\frac{d}{dT} A(T) = -(A(T) - A^*) U, \quad (2)$$

where A^* is the train-loss minimizer and the empirical feature correlation matrix is

$$U = \frac{1}{n} \sum_{\mu=1}^n \mathbb{E}_{\xi} [\phi(x_t^{\mu}) \phi(x_t^{\mu})^{\top}], \quad (3)$$

$$\phi(x_t) := \frac{1}{\sqrt{p}} \tanh\left(\frac{1}{\sqrt{d_{\text{latent}}}} W x_t\right), \quad (4)$$

with x_t^{μ} the noisy diffused state of training point μ and ξ the diffusion noise. Diagonalizing Eq. (2) in the eigenbasis of U gives an exponential per-mode evolution

$$a_i(T) - a_i^* = (a_i(0) - a_i^*) e^{-\lambda_i T}, \quad (5)$$

so eigenmode i of U is absorbed by the readout with characteristic timescale $\tau_i = 1/\lambda_i$.

Estimating U . U is computed numerically per run: we fix the training set and the random first layer W , draw a Monte Carlo estimate of the expectation over diffusion noise ξ in Eq. (3) (500 Monte Carlo noise samples; Appendix A),

assemble the $p \times p$ matrix, and then take its eigendecomposition. The reported spectra are the eigenvalues of this empirical estimate; no analytic surrogate is used.

What the eigenvalues mean. Eq. (5) says each λ_i is a learning rate for one feature direction: large λ_i are absorbed quickly, small ones slowly, and the order in which the readout fits its mass is exactly the sorted eigenvalue list. The RFNN is therefore performing weighted PCA on a frozen nonlinear basis: the random map ϕ defines a fixed set of candidate directions in $\mathbb{R}^p$, U is the empirical covariance of the training data after that lift, and λ_i measures how strongly the training samples vote for direction i . Modes are learned in order of how well the data supports them. Our extension operates entirely inside this framework; only the data covariance Σ_{data} changes.

3.2. From two bulks to four

Bonnaire et al. prove (replica limit, isotropic $\Sigma_{\text{data}} = \mathbb{I}$) that the spectrum of U splits into a population bulk ρ_2 and a sample bulk ρ_1 . With anisotropic data covariance $\Sigma_{\text{data}} = \text{diag}(\sigma_{\text{sig}}^2 \mathbb{I}_{d_{\text{intrinsic}}}, \sigma_{\perp}^2 \mathbb{I}_{d_{\text{latent}} - d_{\text{intrinsic}}})$ ($\sigma_{\text{sig}} \gg \sigma_{\perp}$), we hypothesize the sorted spectrum instead splits into *four* bulks along the 2×2 axes (shared across all training points vs. unique to individual training points) $\times$ (signal vs. null subspace): (i) BULK_DATA_SIGNAL: meaningful directions that every training point shares – the $d_{\text{intrinsic}}$ -dimensional signal subspace where the clusters actually live. These are the largest eigenvalues and correspond to Bonnaire’s ρ_2 in the isotropic case. (ii) BULK_DATA_NOISE: unmeaningful directions that every training point shares – the $d_{\text{latent}} - d_{\text{intrinsic}}$ null subspace, which carries only the small isotropic variance $\sigma_{\perp}^2$. This is the new bulk; it opens whenever $d_{\text{latent}} > d_{\text{intrinsic}}$ and is what produces the noise-dimension buffer. (iii) BULK_SAMPLE_SIGNAL: directions unique to individual training samples within the signal subspace – the sample-specific fluctuations the readout has to fit individually in order to memorize. (iv) BULK_SAMPLE_NOISE: directions unique to individual training samples in the null subspace – the same kind of sample-specific fluctuation but along null directions. Together (iii) and (iv) make up Bonnaire’s ρ_1 . We observe these four populations in every configuration we ran (Figure 1; sweep-level evidence in Appendix K). The cliff at index $d_{\text{intrinsic}}$ between the two data bulks is sharp at both noise scales; the cliff at d_{latent} between data and sample bulks is sharp at $\sigma_{\perp} = 0.5$ and degenerates to a smooth transition at $\sigma_{\perp} = 0.01$. The names above are bookkeeping for the populations we observe, not predictions from the analysis.

Bonnaire et al. (2025) already note (Figure 4, right panel) that ρ_2 develops internal structure when Σ_{data} has multiple eigenvalues, but at signal-to-null ratio $3\times$ they treat it as a

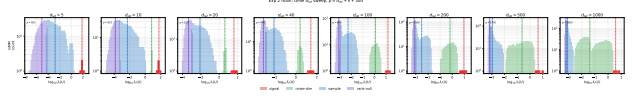


Figure 1. RFNN four-bulk spectrum for the d_{latent} sweep. A synthetic GMM with fixed $d_{\text{intrinsic}} = 5$ is embedded into increasing d_{latent} . The red signal modes stay fixed in count, while the green noise-dimension bulk grows with $d_{\text{latent}} - d_{\text{intrinsic}}$ before the model reaches the blue sample-specific modes.

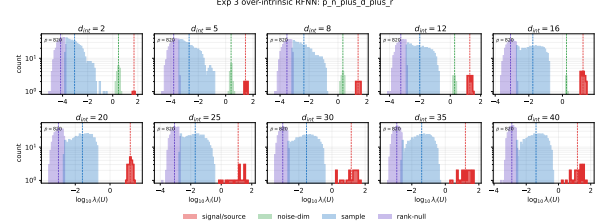


Figure 2. RFNN Exp-3 source-dimensionality sweep. We fix the observed latent space at $d_{\text{latent}} = 20$ and regenerate the source GMM for each $d_{\text{intrinsic}}$, including over-intrinsic cases where the source dimension exceeds the observed latent dimension and is randomly compressed. The signal/source block grows until $d_{\text{intrinsic}} = d_{\text{latent}}$; past that point there is no additional observed signal dimension, so the green noise-dimension buffer disappears.

single deformed bulk. The latent-diffusion regime sits at the opposite extreme ($\sigma_{\text{sig}}^2/\sigma_{\perp}^2 \approx 10^4$ at $\sigma_{\perp} = 0.01$, ≈ 4 at $\sigma_{\perp} = 0.5$): the humps inside ρ_2 separate into well-isolated bulks decades apart, which is why we count them as distinct.

Mechanism: a noise-dimension buffer. The RFNN absorbs modes in order of decreasing λ_i , so the trajectory traverses the four bulks sequentially: signal bulk first (defining τ_{gen} , d_{latent} -independent at fixed σ_{sig} , k , cluster scale); noise-dim bulk next (additional training time, no sample-quality benefit); sample bulk last (defining τ_{mem}). Because the buffer grows as $d_{\text{latent}} - d_{\text{intrinsic}}$, the larger this difference, the longer the time before memorization begins, while τ_{gen} stays fixed. The buffer does not exist in Bonnaire’s isotropic case.

What the gen-gap measures. The training set carries two kinds of structure. The data bulks correspond to *population* structure: directions of variance that exist in the population and happen to also be visible in the training sample. The sample bulk corresponds to *null structure*: directions that look like signal in the n training points but do not exist in the population. The RFNN absorbs eigenmodes in order of decreasing λ_i , so it fits the population structure first and the null structure later. While it is fitting population structure, train and test loss drop together. Once it begins fitting null structure, train loss keeps dropping but test loss does not, and the gap opens. The gen-gap is therefore a direct measure of how much null structure the readout has absorbed, which is why it opens at τ_{mem} and why the noise-dim buffer also delays its onset: widening the buffer pushes back the

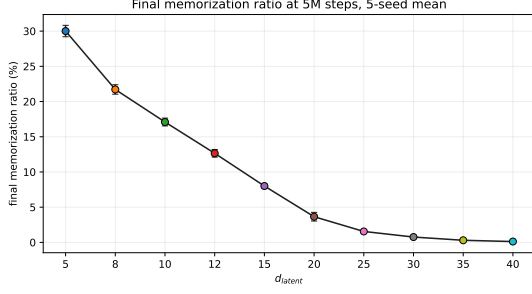


Figure 3. **Synthetic MLP memorization decreases with latent buffer width.** Five-seed mean final memorization ratio after 5M steps for a fixed-width MLP, sweeping d_{latent} at $d_{\text{intrinsic}} = 5$. Memorization falls from roughly 30% at $d_{\text{latent}} = 5$ to near zero by $d_{\text{latent}} = 40$.

moment when null structure begins to be learned.

Decomposing the four-bulk structure. The signal and noise-dim bulks of U inherit the block structure of Σ_{data} directly: $X^\top X/n$ on the same training data has $d_{\text{intrinsic}}$ large eigenvalues in the signal subspace and $d_{\text{latent}} - d_{\text{intrinsic}}$ small eigenvalues in the null directions, separated by roughly $\sigma_{\text{sig}}^2/\sigma_{\perp}^2$, and the lift by W carries them into the data bulks of U up to a constant from W 's spectrum. The sample and rank-null bulks are feature-space artifacts of n training points lifted into $p > d_{\text{latent}}$ random features. The buffer mechanism is thus gradient flow following the data's own variance hierarchy.

4. Synthetic and real-data MLP experiments

4.1. Synthetic MLP

The RFNN microscope predicts that widening the latent space adds low-variance buffer modes before sample-specific modes can be fit. We next test the same mechanism in a trainable score network. The synthetic MLP uses the same Gaussian-mixture data as the RFNN experiment, but the first layer is learned and the score is trained over diffusion times. To avoid the most obvious confound, we hold the hidden width fixed across the d_{latent} sweep; increasing d_{latent} therefore changes the latent buffer and output dimension, but not the hidden-layer capacity.

The MLP result is qualitatively sharper than the frozen-feature prediction: at high d_{latent} , feature learning appears to find sparser, signal-aligned directions instead of spending equal effort on all random-feature modes. Nevertheless, the direction of the effect agrees with the buffer picture: widening the latent space does not accelerate memorization, and under fixed hidden width it strongly suppresses it.



Figure 4. **Synthetic memorization trajectories.** The same five-seed sweep over training steps. Larger latent spaces delay and suppress the onset of memorization under the Bonnaire/Somepalli nearest-neighbor ratio test.

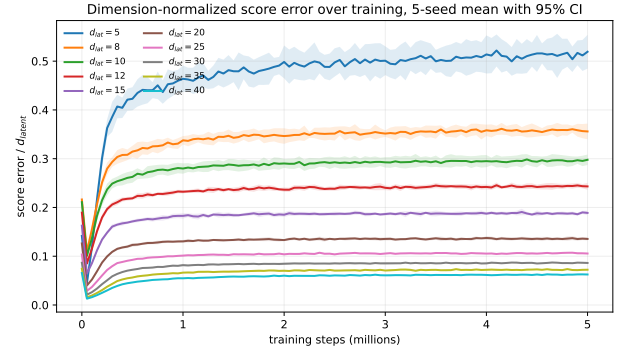


Figure 5. **Synthetic score error per latent dimension.** Normalizing score error by d_{latent} separates the memorization trend from the trivial increase in output dimension. The fixed-width MLP remains comparable across the sweep while memorization drops sharply.

4.2. CelebA and CIFAR-10

Real image latents introduce a second effect absent from the synthetic construction: if the VAE bottleneck is too narrow, the training images cannot be reconstructed faithfully, so pixel-space memorization is hard to detect. We therefore expect a non-monotone curve. As d_{latent} first increases, the VAE becomes expressive enough for the diffusion model to reproduce the 1k-image training subset; once the latent code is sufficiently expressive, further increasing d_{latent} creates an excess-dimensional buffer and memorization decreases.

On CelebA, final memorization peaks near $d_{\text{latent}} = 70$ –80 and then falls by roughly half by $d_{\text{latent}} = 200$. On CIFAR-10, the peak occurs near $d_{\text{latent}} = 100$ and falls to about 12% by $d_{\text{latent}} = 260$. FID shows the corresponding reconstruction/generation tradeoff: moderate dimensions improve quality, while very high dimensions make training harder and degrade FID.

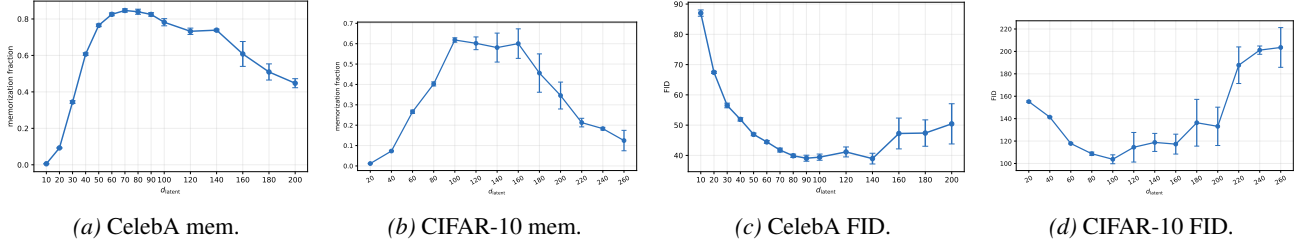


Figure 6. Real-data latent width produces a quality-memory tradeoff. Five-seed means with 95% confidence intervals after 5M MLP diffusion steps on a diverse 1k subset. On both datasets, memorization rises at small-to-moderate d_{latent} as the VAE becomes more faithful, then decreases at larger d_{latent} . FID is best at intermediate latent widths and worsens again once the latent space is very wide.

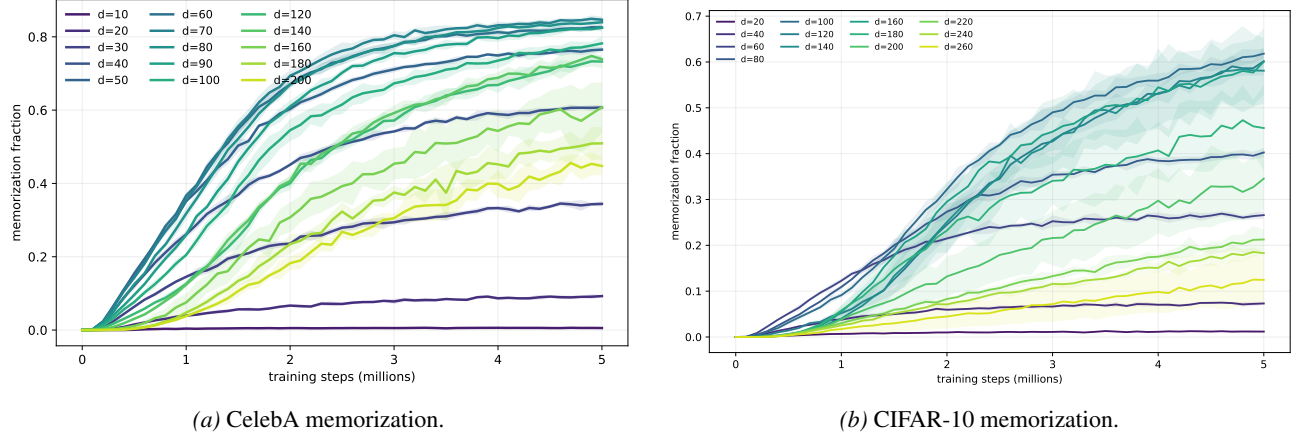


Figure 7. Memorization over training. Mean memorization fraction over five seeds. High-dimensional CelebA and CIFAR-10 runs remain below the mid-dimensional peak throughout training, rather than merely reaching the same endpoint later.

4.3. Data-spectrum predictor

Finally, we test whether the empirical memorization curves are predictable from the data geometry alone. For each trained VAE we encode the same diverse 1k subset, form the diffused latent covariance spectrum, and compute a spectral pressure curve $P_d(s)$ that accumulates learned eigenmodes over training step s . A single global calibration maps pressure levels to memorization thresholds. This is not yet the closed-form theorem; it is a data-spectrum surrogate that asks whether the geometry of the latent code contains the right temporal ordering.

5. Conclusion and limitations

Conclusion. The experiments support a simple mechanistic picture: latent dimensions beyond the data’s effective signal dimension are not just wasted coordinates. In the RFNN they appear as a distinct noise-dimension bulk between population and sample-specific modes. In the trainable synthetic MLP, increasing the latent buffer sharply reduces memorization under fixed hidden width. In real VAE latents, the same force competes with VAE fidelity: small bottlenecks do not reconstruct well enough to memorize in pixel space, intermediate dimensions both reconstruct and

memorize, and very wide latents again reduce memorization while often hurting FID.

Practical implication. The best latent dimension is not the smallest possible code or the largest possible code. Too small loses reconstruction fidelity; too large makes diffusion harder. The useful regime is near the point where VAE quality has saturated but before the diffusion model pays too much optimization cost. The spectral predictor in Sec. 4.3 is a first attempt to estimate this regime directly from the latent data spectrum, without running every diffusion training curve to completion.

Limitations. (i) The RFNN theory is still incomplete: the current manuscript has a mechanism and empirical eigenspectra, but the closed-form anisotropic bulk-edge theorem is left as a TODO. (ii) The synthetic data manifold is a linear Gaussian mixture, so curved manifolds and class hierarchies are only tested indirectly through CelebA and CIFAR-10. (iii) The real-data memorization metric is pixel-space nearest-neighbor ratio on a diverse 1k subset; other definitions may shift the absolute percentages. (iv) The VAE and diffusion systems interact: low-dimensional VAEs suppress measured memorization by blurring reconstructions, while high-dimensional VAEs can degrade

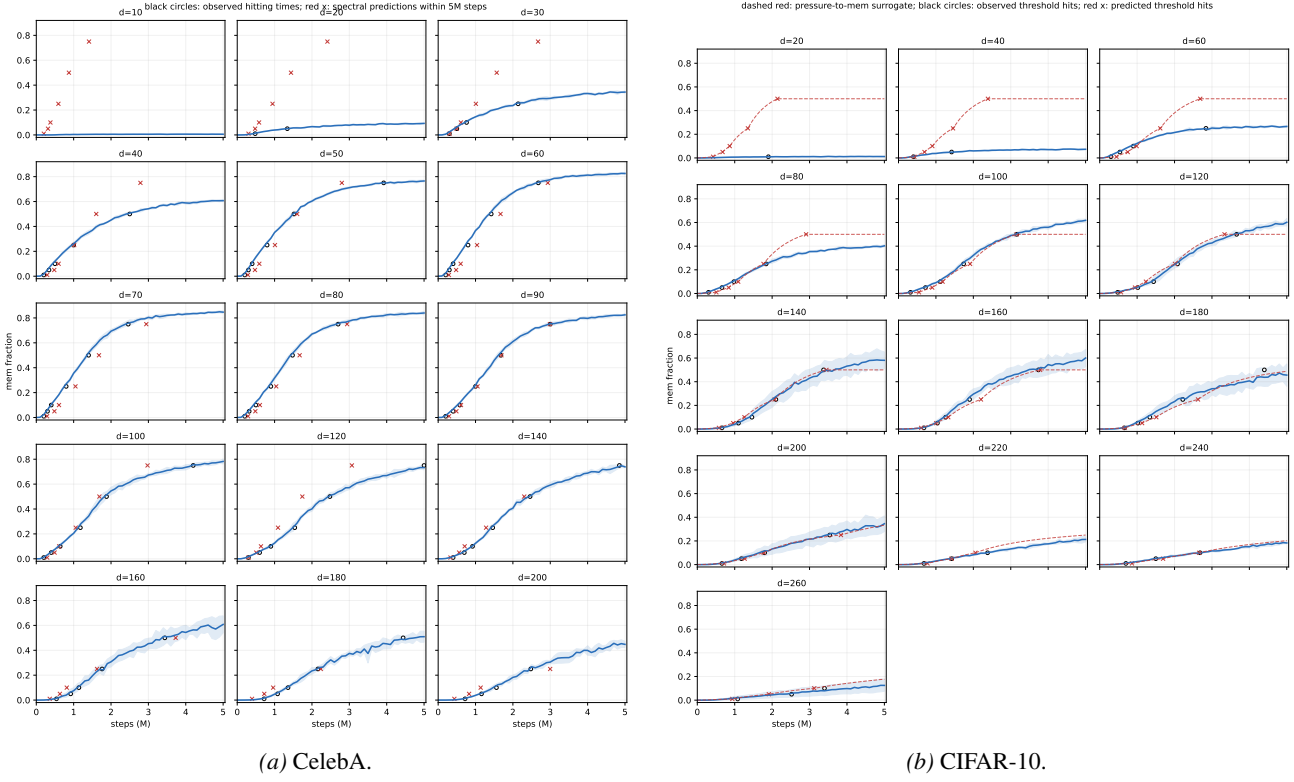


Figure 8. Spectral memorization predictor. Empirical memorization curves with predicted iso-memorization checkpoints overlaid. The calibration is global across dimensions within a dataset; no per- d fit is used. The strongest match is the ordering and onset pattern across latent widths, which is the quantity needed for a stopping-time heuristic.

FID through optimization difficulty. Separating these effects more cleanly is the main next ablation.

References

- Benigni, L. and Péché, S. Eigenvalue distribution of some nonlinear models of random matrices. *Electronic Journal of Probability*, 26(none):1 – 37, 2021. doi: 10.1214/21-EJP699.
- Bonnaire, T., Urfin, R., Biroli, G., and Mezard, M. Why diffusion models don’t memorize: The role of implicit dynamical regularization in training. In Belgrave, D., Zhang, C., Lin, H., Pascanu, R., Koniusz, P., Ghassemi, M., and Chen, N. (eds.), *Advances in Neural Information Processing Systems*, volume 38, pp. 141266–141286. Curran Associates, Inc., 2025.
- Brooks, T., Peebles, B., Holmes, C., et al. Video generation models as world simulators. OpenAI technical report, 2024.
- Brown, B. C., Caterini, A. L., Ross, B. L., Cresswell, J. C., and Loaiza-Ganem, G. Verifying the union of manifolds hypothesis for image data. In *The Eleventh International Conference on Learning Representations*, 2023.
- Carlini, N., Hayes, J., Nasr, M., Jagielski, M., Sehwag, V., Tramèr, F., Balle, B., Ippolito, D., and Wallace, E. Extracting training data from diffusion models. In *32nd USENIX Security Symposium (USENIX Security 23)*, pp. 5253–5270, Anaheim, CA, August 2023. USENIX Association. ISBN 978-1-939133-37-3.
- Couillet, R. and Debbah, M. *Random Matrix Methods for Wireless Communications*. Cambridge University Press, 2011.
- Esser, P., Kulal, S., Blattmann, A., Entezari, R., Müller, J., Saini, H., Levi, Y., Lorenz, D., Sauer, A., Boesel, F., Podell, D., Dockhorn, T., English, Z., and Rombach, R. Scaling rectified flow transformers for high-resolution image synthesis. In Salakhutdinov, R., Kolter, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and Berkenkamp, F. (eds.), *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 12606–12633. PMLR, 21–27 Jul 2024.
- Facco, E., d’Errico, M., Rodriguez, A., and Laio, A. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports*, 7(1): 12140, 2017. doi: 10.1038/s41598-017-11873-y.
- Gu, X., Du, C., Pang, T., Li, C., Lin, M., and Wang, Y. On memorization in diffusion models. *Transactions on Machine Learning Research*, 2025. ISSN 2835-8856.
- Ho, J., Jain, A., and Abbeel, P. Denoising diffusion probabilistic models. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H. (eds.), *Advances in Neural Information Processing Systems*, volume 33, pp. 6840–6851. Curran Associates, Inc., 2020.
- Jacot, A., Gabriel, F., and Hongler, C. Neural tangent kernel: Convergence and generalization in neural networks. In Bengio, S., Wallach, H., Larochelle, H., Grauman, K., Cesa-Bianchi, N., and Garnett, R. (eds.), *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018.
- Kadkhodaie, Z., Guth, F., Simoncelli, E. P., and Mallat, S. Generalization in diffusion models arises from geometry-adaptive harmonic representations. In *The Twelfth International Conference on Learning Representations*, 2024.
- Kamkari, H., Ross, B. L., Hosseinzadeh, R., Cresswell, J., and Loaiza-Ganem, G. A geometric view of data complexity: Efficient local intrinsic dimension estimation with diffusion models. In Globerson, A., Mackey, L., Belgrave, D., Fan, A., Paquet, U., Tomczak, J., and Zhang, C. (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 38307–38354. Curran Associates, Inc., 2024. doi: 10.52202/079017-1211.
- Karoui, N. E. The spectrum of kernel random matrices. *The Annals of Statistics*, 38(1):1 – 50, 2010. doi: 10.1214/08-AOS648.
- Karras, T., Aittala, M., Aila, T., and Laine, S. Elucidating the design space of diffusion-based generative models. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A. (eds.), *Advances in Neural Information Processing Systems*, volume 35, pp. 26565–26577. Curran Associates, Inc., 2022.
- Lee, S., Kreis, K., Veccham, S. P., Liu, M., Reidenbach, D., Peng, Y., Paliwal, S. G., Nie, W., and Vahdat, A. Genmol: A drug discovery generalist with discrete diffusion. In *Forty-second International Conference on Machine Learning*, 2025.
- Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M., and Le, M. Flow matching for generative modeling. In *The Eleventh International Conference on Learning Representations*, 2023.
- Mei, S., Montanari, A., and Nguyen, P.-M. A mean field view of the landscape of two-layer neural networks. *Proceedings of the National Academy of Sciences*, 115(33): E7665–E7671, 2018. doi: 10.1073/pnas.1806579115.
- Pennington, J. and Worah, P. Nonlinear random matrix theory for deep learning. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S., and Garnett, R. (eds.), *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017.

- Podell, D., English, Z., Lacey, K., Blattmann, A., Dockhorn, T., Müller, J., Penna, J., and Rombach, R. SDXL: Improving latent diffusion models for high-resolution image synthesis. In *The Twelfth International Conference on Learning Representations*, 2024.
- Pope, P., Zhu, C., Abdelkader, A., Goldblum, M., and Goldstein, T. The intrinsic dimension of images and its impact on learning. In *International Conference on Learning Representations*, 2021.
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., and Ommer, B. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10684–10695, June 2022.
- Somepalli, G., Singla, V., Goldblum, M., Geiping, J., and Goldstein, T. Diffusion art or digital forgery? investigating data replication in diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 6048–6058, 2023.
- Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., and Poole, B. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.
- Stanczuk, J. P., Batzolis, G., Deveney, T., and Schönlieb, C.-B. Diffusion models encode the intrinsic dimension of data manifolds. In Salakhutdinov, R., Kolter, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and Berkenkamp, F. (eds.), *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 46412–46440. PMLR, 21–27 Jul 2024.

A. Experimental methods and design choices

A.1. Data generation

We generate anisotropic Gaussian-mixture data:

- Cluster centers $\{m_c\}_{c=1}^k$ are drawn iid uniformly on a sphere of radius s inside the first $d_{\text{intrinsic}}$ coordinates of $\mathbb{R}^{d_{\text{latent}}}$; the remaining $d_{\text{latent}} - d_{\text{intrinsic}}$ “null-space” coordinates of every center are exactly zero.
- For each training point μ , a cluster index $c(\mu)$ is drawn uniformly, then $x^\mu = m_{c(\mu)} + Q \xi^\mu$ where ξ has independent components: variance $\sigma_{\text{sig}}^2 = 1$ along the $d_{\text{intrinsic}}$ signal dimensions and variance $\sigma_{\perp}^2$ along the $d_{\text{latent}} - d_{\text{intrinsic}}$ null-space dimensions. Q is a fixed orthogonal basis whose first $d_{\text{intrinsic}}$ columns span the signal subspace.
- Default values: $k = 10$, $d_{\text{intrinsic}} = 5$, $s = 3$, $n = 500$ training points, 2048 held-out test points, $\sigma_{\text{sig}} = 1$, $\sigma_{\perp} \in \{0.01, 0.5\}$.

A.2. Diffusion process

We use the variance-preserving Ornstein-Uhlenbeck process with $t \in [t_{\min}, t_{\max}] = [0.01, 3.0]$. Score targets are computed analytically from the closed-form Gaussian-mixture density at noise level t ; this avoids the standard score-matching MSE loss noise floor and gives a clean “score error” metric independent of training-set fluctuations.

A.3. Models

MLP. 3-layer ReLU MLP with sinusoidal time embedding (32 frequency bands), input width $d_{\text{latent}} + 64$, output width d_{latent} . Hidden width $h = 256$ held fixed across the d_{latent} sweep. Adam with learning rate 10^{-4} , batch size 256, $T = 3 \times 10^5$ training steps (extended to 5×10^6 for the body timescale-scatter figure), evaluation every 5,000 steps with 5,000 generated samples.

RFNN. Two-layer RFNN with tanh activation; first layer $W \in \mathbb{R}^{p \times d_{\text{latent}}}$ is fixed at random Gaussian initialization, the second-layer readout A is trained. $p = 64 d_{\text{latent}}$ to keep the rank-null bulk visible at every d_{latent} . Same optimizer, batch, and evaluation cadence as the MLP.

A.4. Evaluation metrics

- **Score error:** integrated mean-squared error of the learned score against the analytic Gaussian-mixture score, averaged over 5,000 test points.
- **Memorization fraction:** fraction of generated samples whose nearest training-set neighbor is closer than one third of that neighbor’s own nearest training-set neighbor (Somepalli et al., 2023).
- τ_{gen} : first step at which test loss is within 5% of its per-run minimum.
- τ_{mem} : first step at which the memorization fraction exceeds 1%. Censored to “>300k” if not crossed within budget.

Why these definitions. We use the test-loss-plateau criterion for τ_{gen} rather than an FID- or MMD-based criterion because (i) it is well-defined on both the MLP and the RFNN, the latter trained at fixed t with no usable generation process, (ii) the analytic Gaussian-mixture score gives noise-floor-free target values, removing the need for finite-sample FID estimates that fluctuate with the number of generated samples, and (iii) the test-loss plateau aligns directly with the moment when the population score has been internalised. We use the Somepalli et al. (2023)-style memorization fraction for τ_{mem} on the MLP because it is the same threshold Bonnaire et al. (2025) use, allowing direct comparison with the isotropic baseline. For the fixed- t RFNN, no generation is available; we substitute the gen-gap proxy (test loss – train loss > 0.02), which monotonically tracks the canonical τ_{mem} on the MLP and calibrates the two-track comparison.

A.5. Per-experiment configurations

Both sweeps (Section 2, “Two complementary sweeps”) run on the synthetic Gaussian-mixture data described above. Shared constants across every run: $n = 500$ training points, 2,048 held-out test points, $k = 10$ cluster centers, $s = 3$, $\sigma_{\text{sig}} = 1$, OU diffusion at $t \in [0.01, 3.0]$, seed 42 for both data and the random first layer W . Tables 1 and 2 list the sweeps and training hyperparameters.

Table 1. Sweep configurations. Both $\sigma_{\perp}$ values used unless noted.

Model	Vary	Range	Fixed	Width
MLP	d_{latent}	{5, 8, 10, 15, 20, 30, 40}	$d_{\text{intrinsic}} = 5$	$h = 256$
RFNN	d_{latent}	{5, 8, 10, 15, 20, 30, 40}	$d_{\text{intrinsic}} = 5$	$p = 64 d_{\text{latent}}$
RFNN (ext., $\sigma_{\perp}=0.5$)	d_{latent}	{60, 80, 100, 150, 200}	$d_{\text{intrinsic}} = 5$	$p = 64 d_{\text{latent}}$
MLP	$d_{\text{intrinsic}}$	{2, 5, 8, 12, 16, 20}	$d_{\text{latent}} = 20$	$h = 256$
RFNN	$d_{\text{intrinsic}}$	{2, 5, 8, 12, 16, 20}	$d_{\text{latent}} = 20$	$p = 1280$

Table 2. Training hyperparameters.

	MLP	RFNN
Optimizer	Adam	full-batch GD
Learning rate	10^{-4}	closed-form
Batch size	256	full ($n = 500$)
Total steps T	3×10^5	3×10^5
Eval interval	5,000	5,000
Diffusion time	$t \sim \text{Unif}[0.01, 3.0]$	fixed $t = 0.01$
Time embedding	sinusoidal, 32 frequencies	n/a
Activation	ReLU	tanh
U Monte Carlo	n/a	50 noise samples per training point

Table 3. Empirical bulk sizes per d_{latent} . B4 collapses to $d_{\text{intrinsic}} = 5$ as predicted; B3 grows linearly with $d_{\text{latent}} - d_{\text{intrinsic}}$ (the noise-dim buffer); B1 absorbs the rank-null tail.

d_{latent}	B1	B2	B3	B4
5	150	138	27	5
8	344	133	30	5
10	295	305	35	5
15	699	216	40	5
20	916	311	48	5
30	1397	458	60	5
40	1907	578	70	5

A.6. Empirical bulk sizes

Table 3 reports the per-panel sizes of the four detected bulks in Figure 30, ordered left-to-right by increasing eigenvalue (B1: rank-null tail; B4: signal block). $\sigma_{\perp} = 0.01$, $d_{\text{intrinsic}} = 5$, $n = 500$, $p = 64 d_{\text{latent}}$.

A.7. Known pitfalls

The tanh activation in the RFNN saturates whenever the data scale divided by $\sqrt{d_{\text{latent}}}$ exceeds approximately 2, collapsing all eigenvalues into a degenerate single bulk. Configurations were chosen to keep the operating point well inside the linear regime; see Appendix I.

B. d_{latent} sweep (synthetic data)

The body reports the updated five-seed d_{latent} sweep at fixed $d_{\text{intrinsic}} = 5$ (Figs. 3–5). This appendix preserves the older synthetic diagnostic views: timescales, late gen-gap, and score-error curves for both the d_{latent} and $d_{\text{intrinsic}}$ sweeps.

C. d_{latent} sweep at $\sigma_{\perp} = 0.01$

The $\sigma_{\perp} = 0.01$ counterparts of the d_{latent} sweep figures in Appendix B. The non-monotonic score-error peak discussed in Sec. 5 is visible in Fig. 17.

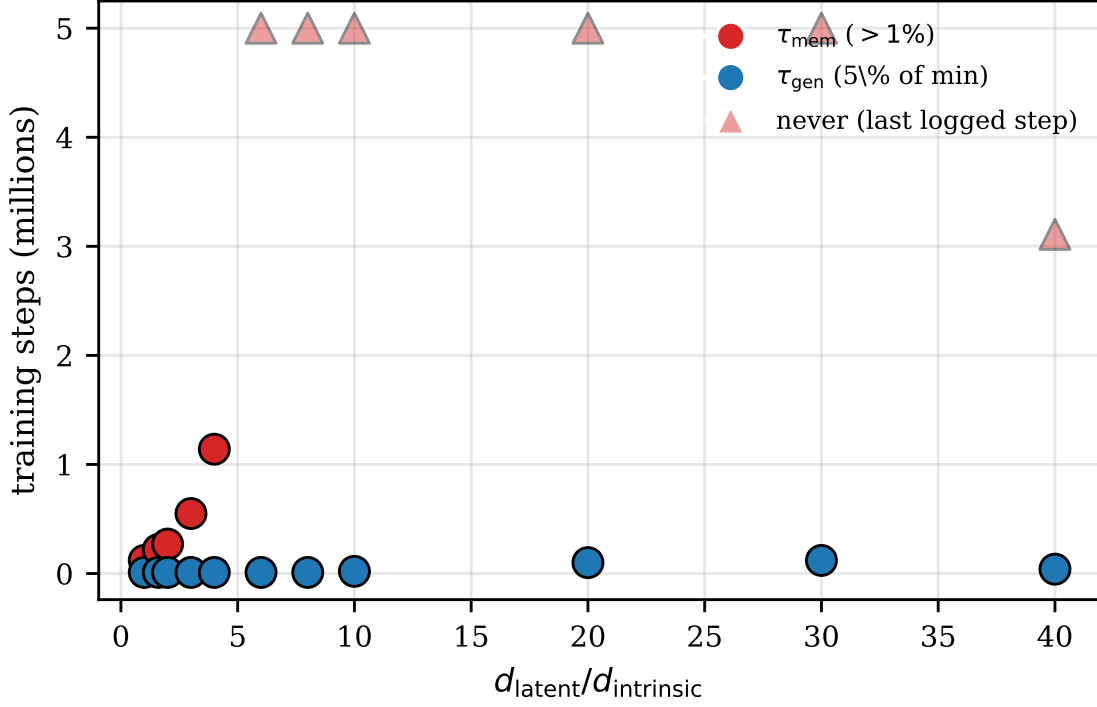


Figure 9. d_{latent} sweep: timescales versus $d_{\text{latent}}/d_{\text{intrinsic}}$ at $\sigma_{\perp} = 0.5$. 5M-step MLP. Triangles mark runs that do not reach 1% memorization within budget.

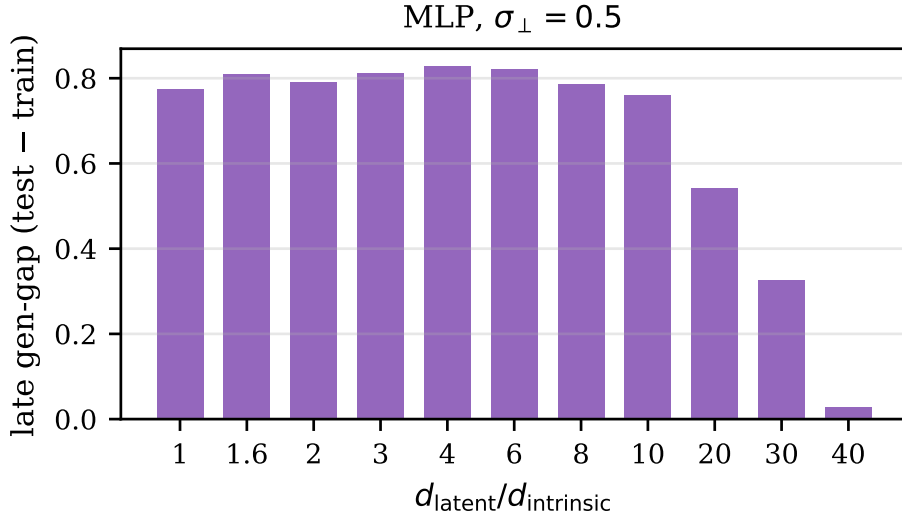


Figure 10. d_{latent} sweep: late generalization gap versus $d_{\text{latent}}/d_{\text{intrinsic}}$ at $\sigma_{\perp} = 0.5$.

D. $d_{\text{intrinsic}}$ sweep additional figures

This section collects additional $d_{\text{intrinsic}}$ -sweep diagnostics at fixed $d_{\text{latent}} = 20$, complementary to the updated d_{latent} -sweep MLP figures in the body.

E. $d_{\text{intrinsic}}$ sweep at $\sigma_{\perp} = 0.01$

The $\sigma_{\perp} = 0.01$ counterparts of the body's $d_{\text{intrinsic}}$ sweep at fixed $d_{\text{latent}} = 20$. These runs are at 300k training steps (the 5M extension was only run at $\sigma_{\perp} = 0.5$).

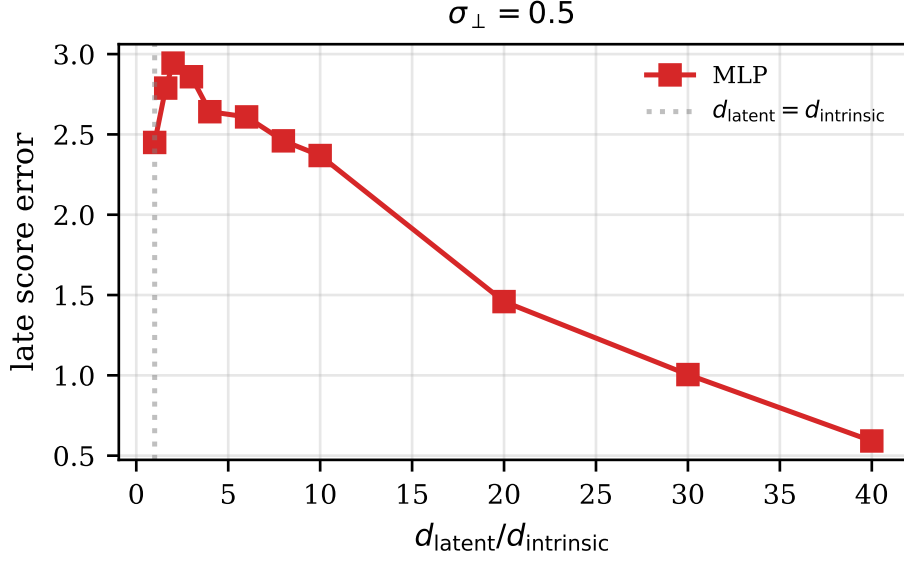


Figure 11. d_{latent} sweep: late score error versus $d_{\text{latent}}/d_{\text{intrinsic}}$ at $\sigma_{\perp} = 0.5$. Compare with the updated normalized score-error trajectories in Fig. 5.

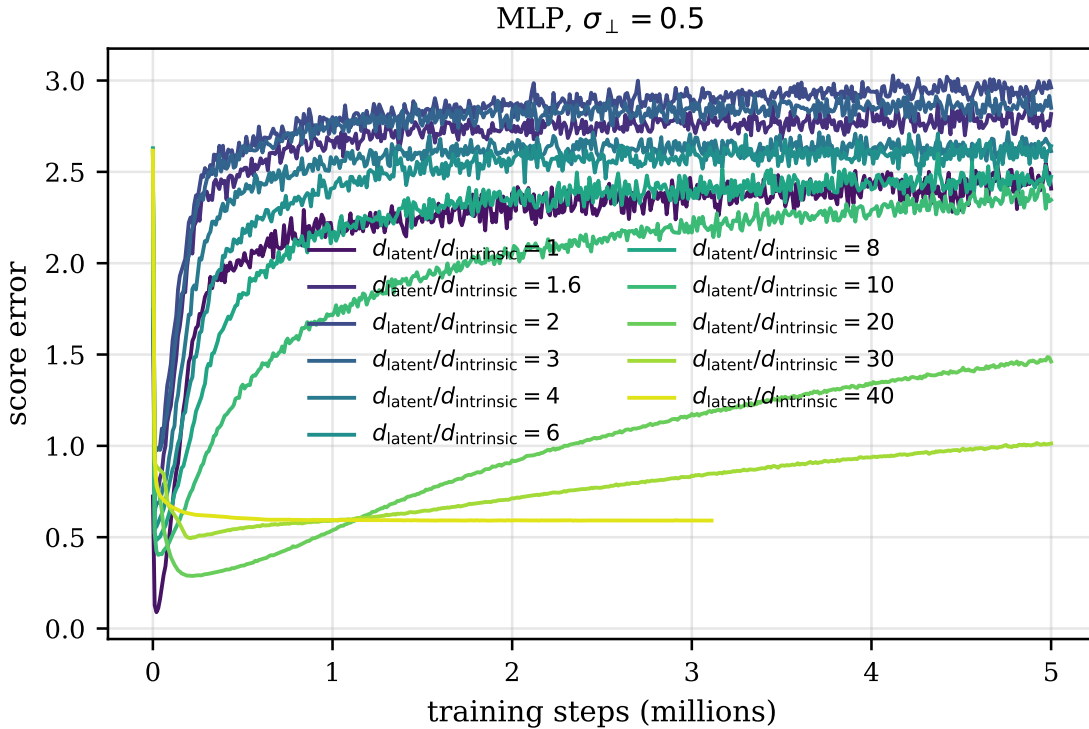


Figure 12. d_{latent} sweep: score error over training at $\sigma_{\perp} = 0.5$. Per-step trajectories underlying the aggregate in Fig. 11, one curve per d_{latent} (viridis colormap, low to high). Compare to Fig. 18 (the $d_{\text{intrinsic}}$ -sweep counterpart).

F. Real-data experiments

F.1. Setup

We test the buffer mechanism on CelebA and CIFAR-10, with the latent space learned by a VAE. For each dataset and latent dimension, the VAE is trained on the full available training split. The diffusion model is then trained on a diverse 1k-image subset of the encoded training set. The score model is the fixed-width MLP from Sec. 4: 3-layer ReLU network, sinusoidal

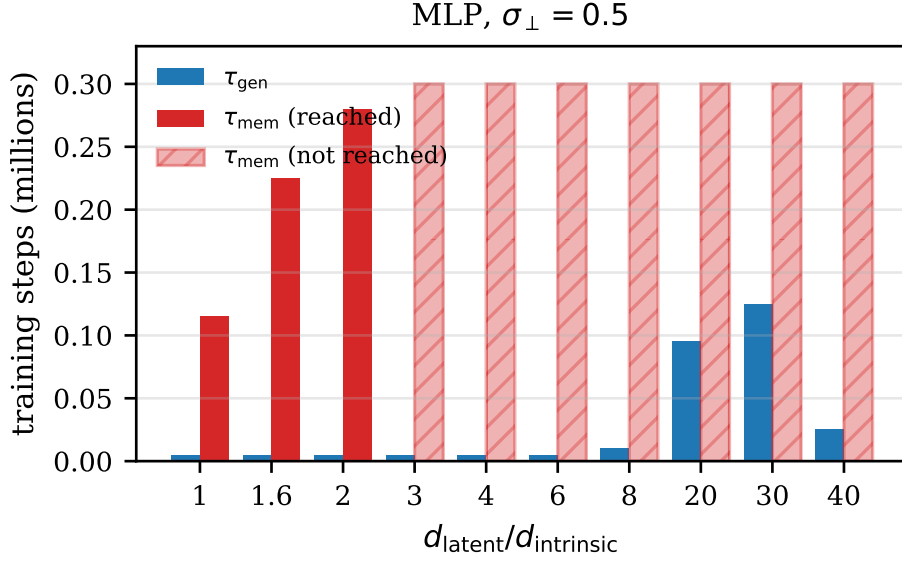


Figure 13. d_{latent} sweep: 300k-step bar-chart timescales at $\sigma_{\perp} = 0.5$. The 300k-step bar-chart view of paired τ_{gen} (blue) and τ_{mem} (red), with hatched bars indicating τ_{mem} not reached within budget. The 5M-step extended-budget version is Fig. 9 above.

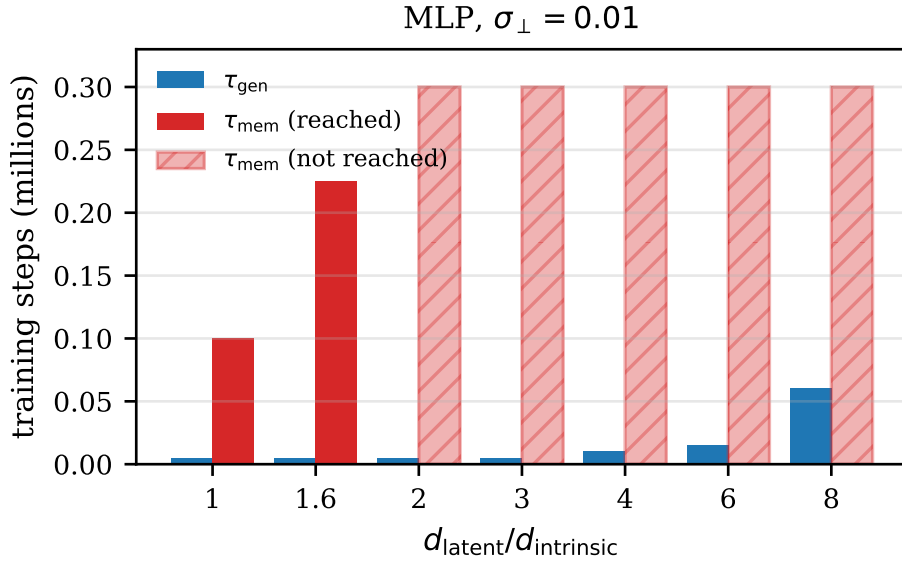


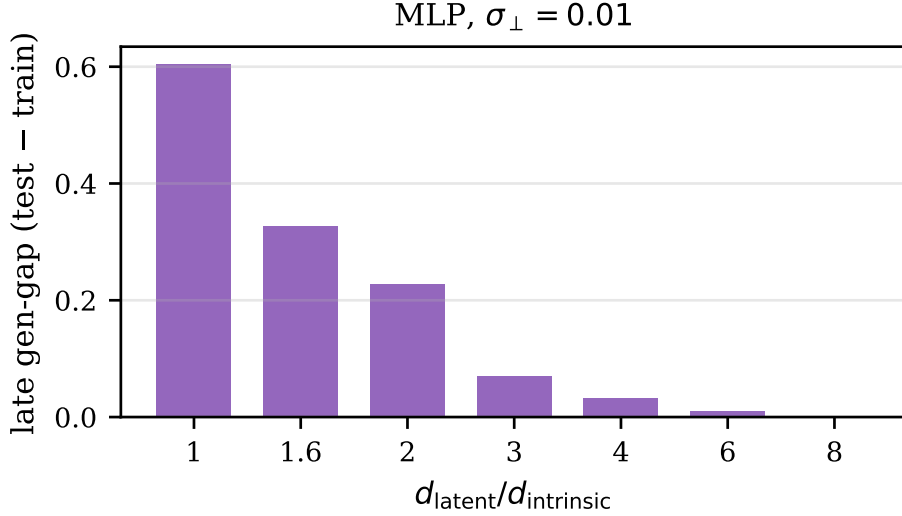
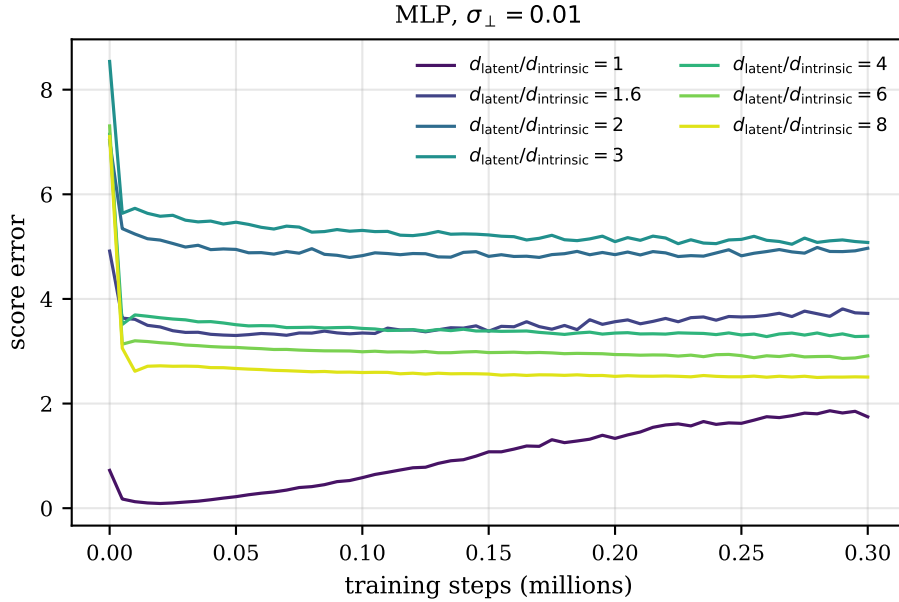
Figure 14. $\sigma_{\perp} = 0.01$ counterpart of Fig. 13.

time embedding, five seeds, 5M training steps, FID evaluated every 100k steps, and pixel-space memorization evaluated by the Bonnaire/Somepalli nearest-neighbor ratio test.

Latent sweeps. CelebA uses $d_{\text{latent}} \in \{10, 20, \dots, 100, 120, 140, 160, 180, 200\}$. CIFAR-10 uses $d_{\text{latent}} \in \{20, 40, \dots, 200, 220, 240, 260\}$. The high-dimensional CIFAR-10 runs exclude exploratory gradient-clipped seeds because clipping substantially suppressed memorization; replacement seeds use the same optimizer and learning rate as the rest of the sweep.

F.2. VAE diagnostics

The VAE itself creates a measurable confound: if the bottleneck is too small, encode–decode reconstructions are too blurred to count as pixel-space memorization. We therefore report VAE validation metrics and reconstruction memorization separately from the diffusion model.


 Figure 15. $\sigma_{\perp} = 0.01$ counterpart of Fig. 10.

 Figure 16. $\sigma_{\perp} = 0.01$ counterpart of Fig. 12.

E.3. Per-dimension diffusion trajectories

E.4. Additional spectral predictor diagnostics

G. RFNN: structural justification of the setup

The RFNN of Eq. (1) is a deliberately constrained model and three of its design choices warrant explanation.

Why freeze the first layer. Freezing W makes the score-matching loss quadratic in the trained readout A , so gradient flow on A is the linear ODE Eq. (2) and admits the closed-form spectral decomposition that underlies every claim in Section 3. A trainable first layer would couple all weights nonlinearly and require either a feature-learning theory (Mei et al., 2018) or a lazy / NTK regime (Jacot et al., 2018); neither delivers the explicit per-mode timescale $\tau_i = 1/\lambda_i$ we use to label τ_{gen} and τ_{mem} . Bonnaire et al. (2025) establish that the frozen-feature RFNN reproduces the qualitative two-bulk generalization/memorization picture observed in trainable networks, so we adopt their setup directly and inherit its

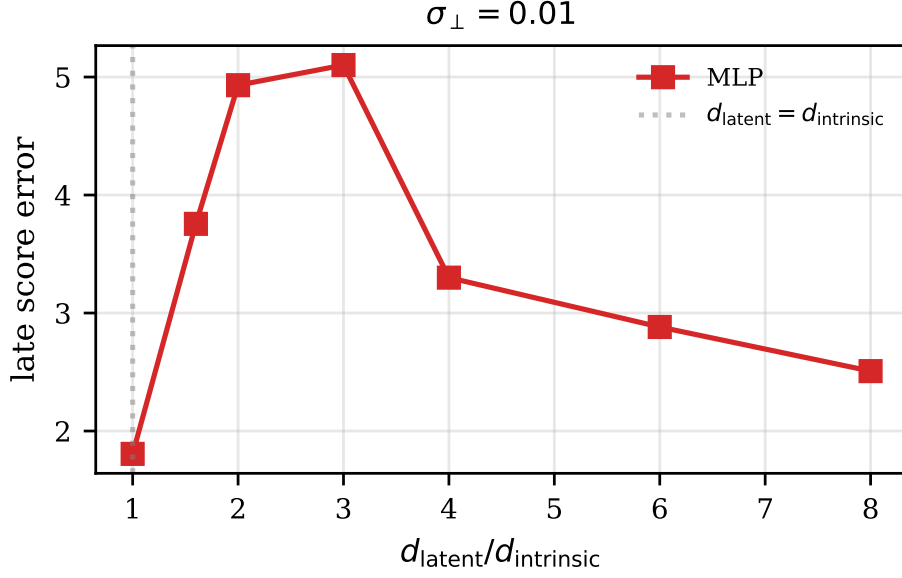


Figure 17. $\sigma_{\perp} = 0.01$ counterpart of Fig. 11. The non-monotonic peak discussed in Sec. 5 is visible here.

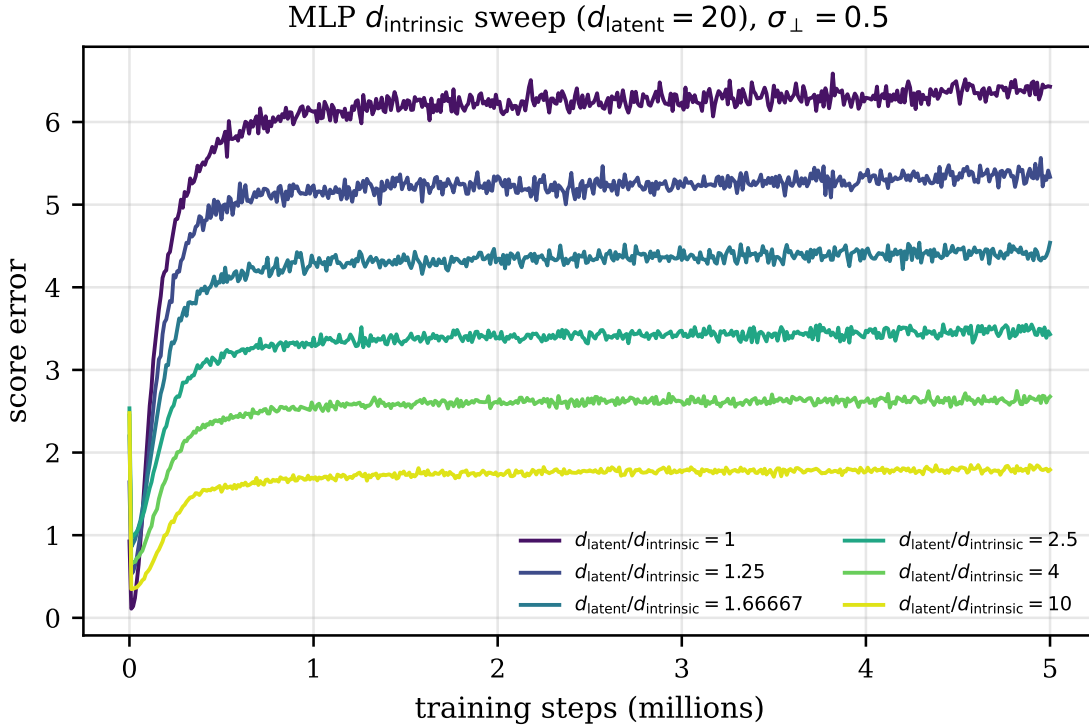


Figure 18. $d_{\text{intrinsic}}$ sweep: score error over training at $\sigma_{\perp} = 0.5$. One curve per $d_{\text{latent}}/d_{\text{intrinsic}}$ ratio (viridis colormap). Per-step trajectories underlying the body’s $d_{\text{intrinsic}}$ -sweep aggregates.

tractability.

Why fix $t = 0.01$. Score matching at a single small diffusion time is the regime in which the population score is sharpest and memorization is most acute (the noise-to-signal ratio is smallest). Training at one fixed t removes the time-integration in standard score matching without changing the spectral structure of $\mathcal{U}$, since $\phi(x_t)$ depends on t only through the diffused state’s covariance. This matches Bonnaire et al.’s setup and isolates the spectral mechanism from time-averaging effects.

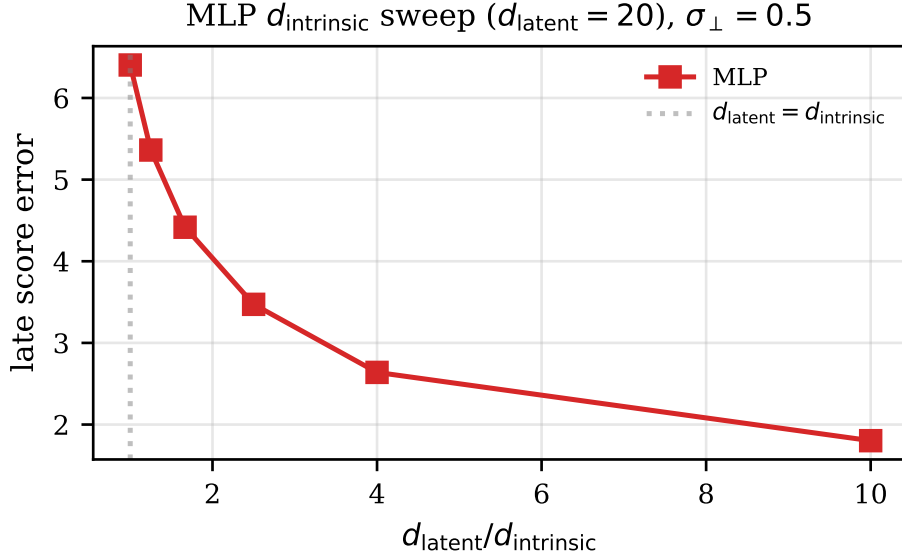


Figure 19. $d_{\text{intrinsic}}$ sweep: late score error versus $d_{\text{latent}}/d_{\text{intrinsic}}$ at $\sigma_{\perp} = 0.5$. $d_{\text{intrinsic}}$ -sweep counterpart of the wider-range d_{latent} sweep.

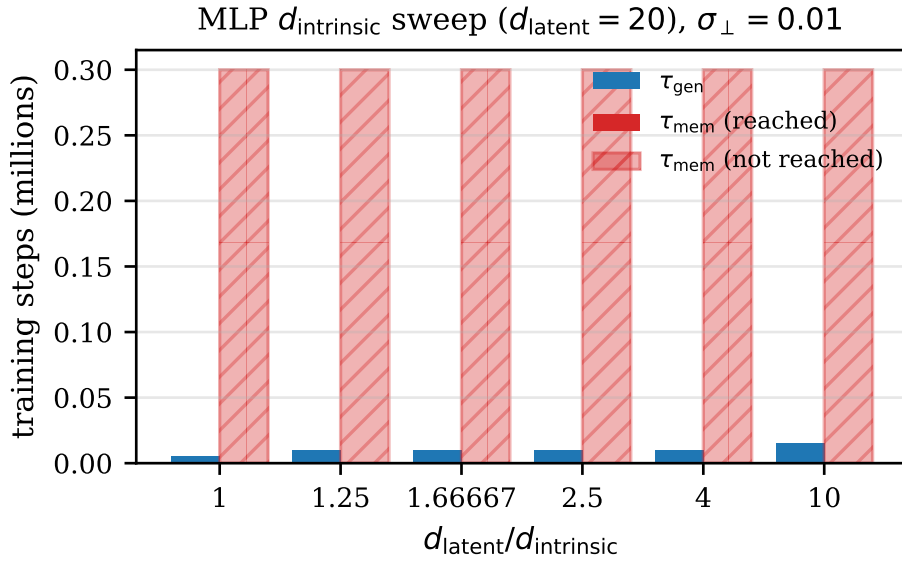
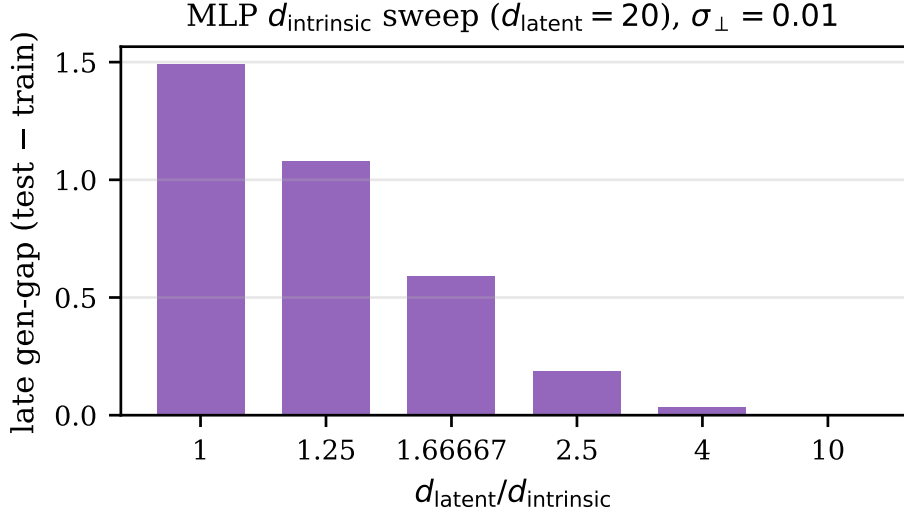
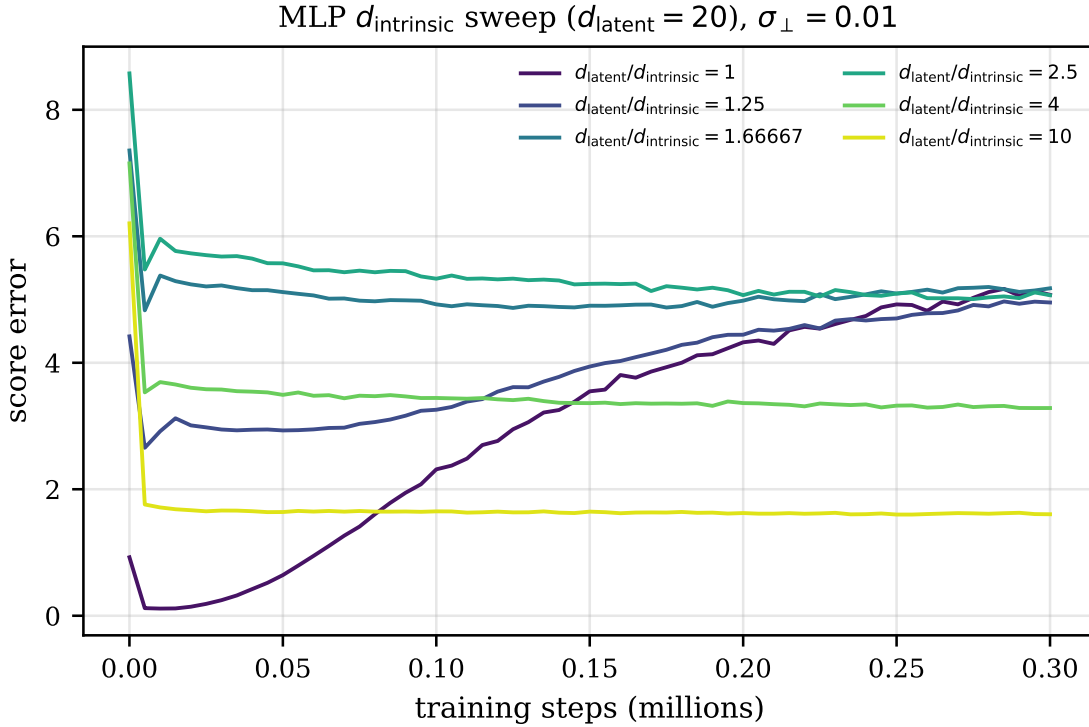


Figure 20. $\sigma_{\perp} = 0.01$ $d_{\text{intrinsic}}$ -sweep timescales.

Why tanh and $\psi_p = p/d_{\text{latent}} = 64$. tanh is the activation Bonnaire et al. (2025) analyze and the saturation pitfall (Appendix I) is well characterized: configurations with $s/\sqrt{d_{\text{latent}}} \lesssim 2$ stay in the regime where the linearised feature map of Appendix H applies. The width $p = 64 d_{\text{latent}}$ is chosen to keep the rank-null bulk visible and the four-bulk structure clearly separated at every d_{latent} ; smaller p collapses the rank-null and sample bulks because $p - d_{\text{latent}} - n$ shrinks, while much larger p provides no additional structural information at fixed n/d_{latent} .

Connection to the trainable MLP. The Section 4 MLP results recover the same buffer mechanism qualitatively (delayed τ_{mem} , monotone late gen-gap), confirming that the RFNN idealization is not a degenerate case of the broader feature-learning picture but a tractable representative of it.


 Figure 21. $\sigma_{\perp} = 0.01$ $d_{\text{intrinsic}}$ -sweep gen-gap.

 Figure 22. $\sigma_{\perp} = 0.01$ counterpart of Fig. 18.

H. RFNN: sketch of an analytical derivation

We sketch the extension of Bonnaire et al.’s replica computation to block-diagonal data covariance $\Sigma_{\text{data}} = \text{diag}(\sigma_{\text{sig}}^2 \mathbb{I}_{d_{\text{intrinsic}}}, \sigma_{\perp}^2 \mathbb{I}_{d_{\text{latent}} - d_{\text{intrinsic}}})$. Throughout, $x^{\mu} = m_{c(\mu)} + \xi^{\mu}$ with $m_{c(\mu)}$ a cluster center in the $d_{\text{intrinsic}}$ -dimensional signal subspace and $\xi^{\mu} \sim \mathcal{N}(0, \Sigma_{\text{data}})$. The diffused state at time t is $x_t^{\mu} = e^{-t} x^{\mu} + \sqrt{\Delta_t} \eta^{\mu}$ with $\Delta_t = 1 - e^{-2t}$ and $\eta^{\mu} \sim \mathcal{N}(0, \mathbb{I}_{d_{\text{latent}}})$.

Linearised feature map. For data scale s small compared with $\sqrt{d_{\text{latent}}}$ (the regime in which our experiments operate; cf. Appendix I), $\tanh(Wx/\sqrt{d_{\text{latent}}}) \approx Wx/\sqrt{d_{\text{latent}}} - \frac{1}{3}(Wx)^3/d_{\text{latent}}^{3/2}$, and to leading order $\phi(x) =$

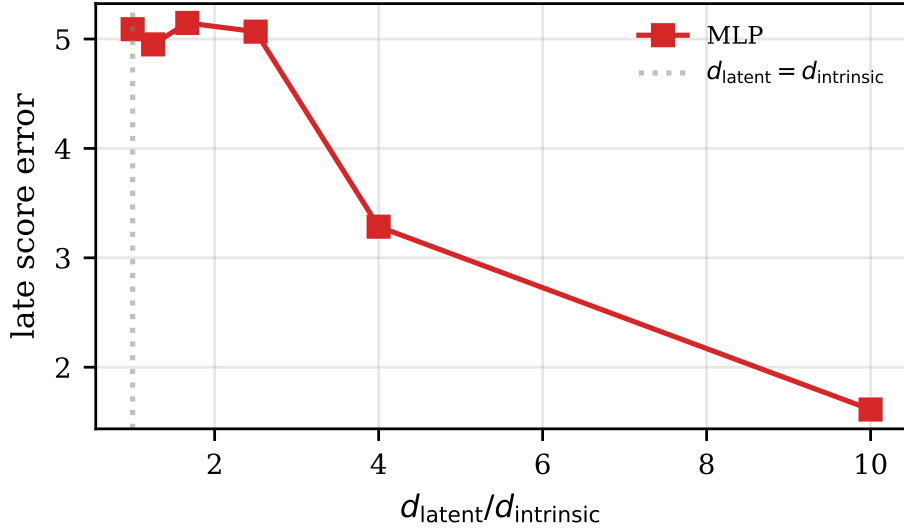
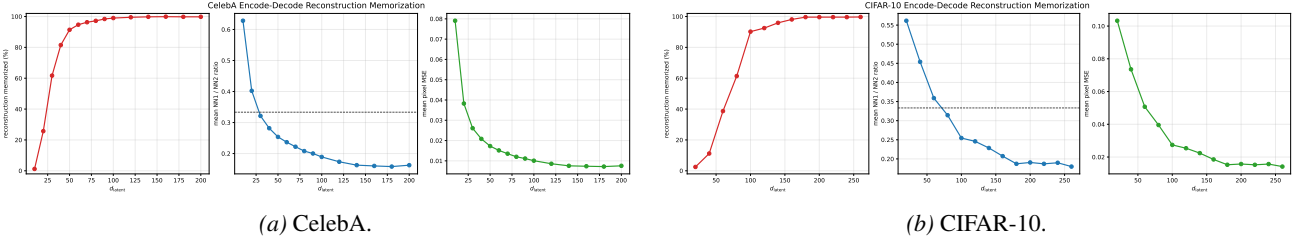
MLP $d_{\text{intrinsic}}$ sweep ($d_{\text{latent}} = 20$), $\sigma_{\perp} = 0.01$

 Figure 23. $\sigma_{\perp} = 0.01$ $d_{\text{intrinsic}}$ -sweep late score error.


Figure 24. **VAE encode-decode reconstruction memorization.** The same Bonnaire/Somepalli nearest-neighbor ratio test is applied to VAE reconstructions of the diverse 1k subset. Low d_{latent} VAEs reconstruct too poorly to trigger the pixel-space memorization test; higher d_{latent} reconstructions increasingly return to their original training examples.

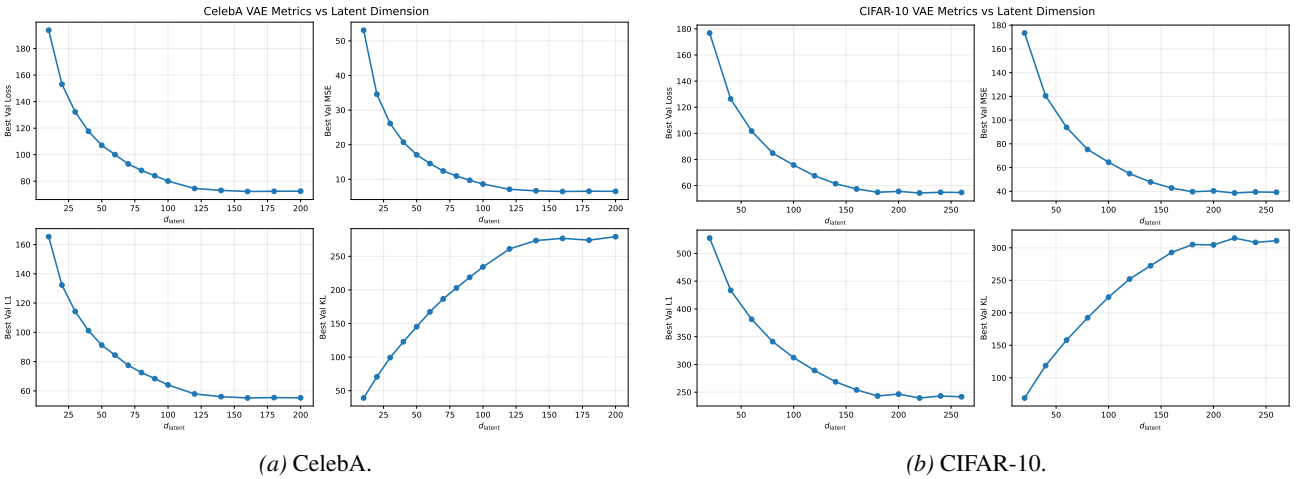


Figure 25. **VAE validation metrics.** Reconstruction loss, MSE, L1, and KL across the latent-dimension sweep. These curves explain the rising side of the real-data memorization plots: before the VAE is expressive enough, the diffusion model cannot reproduce training images sharply in pixel space.

$(1/\sqrt{p}) \tanh(Wx/\sqrt{d_{\text{latent}}}) \approx Wx/\sqrt{pd_{\text{latent}}}$. The linearised correlation matrix is

$$U^{\text{lin}} = \frac{1}{pd_{\text{latent}}} W M_t W^{\top}, \quad M_t := \frac{1}{n} \sum_{\mu} \mathbb{E}_{\eta}[x_t^{\mu} (x_t^{\mu})^{\top}], \quad (6)$$

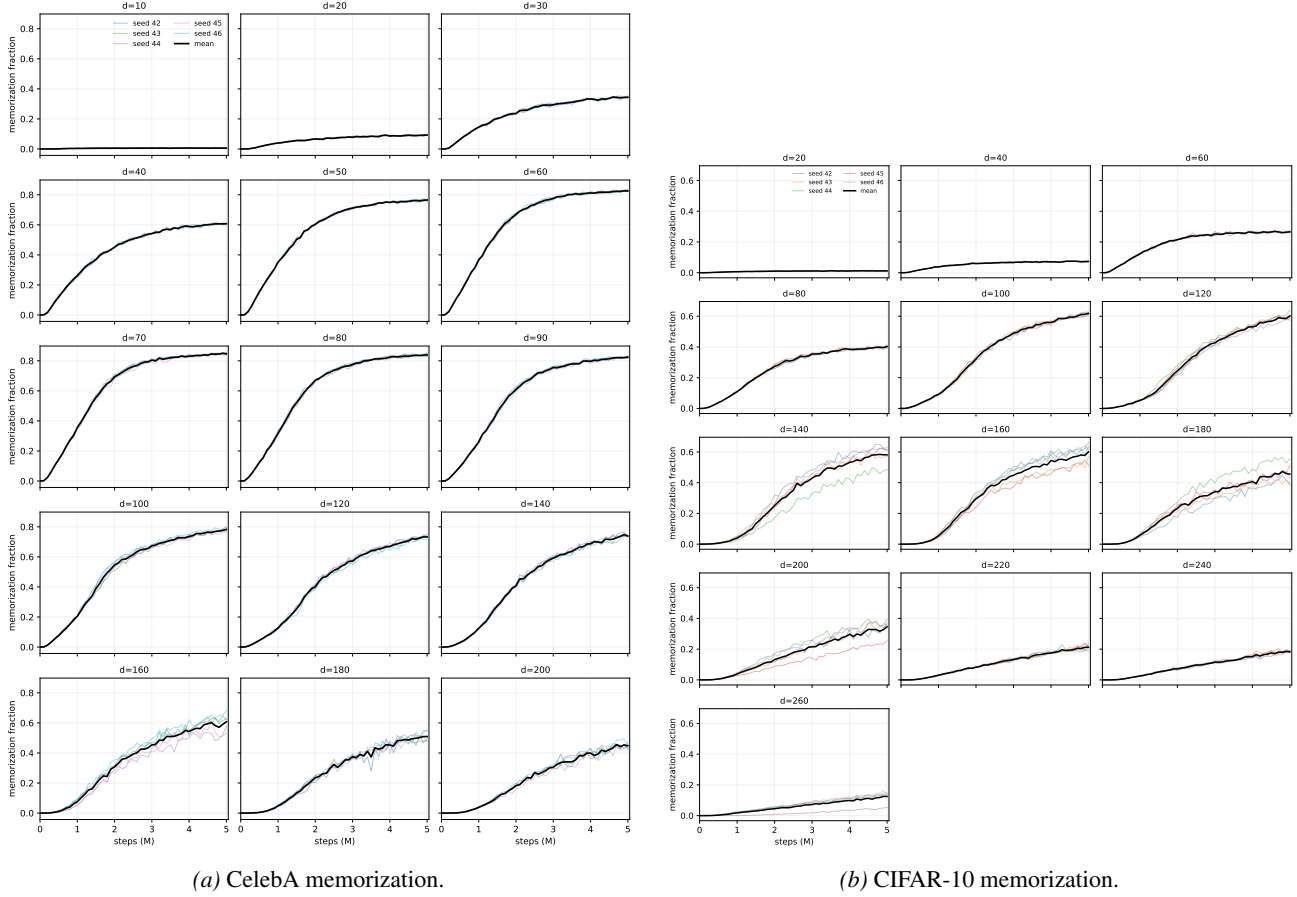


Figure 26. **Per- d_{latent} memorization trajectories.** Each panel fixes one latent dimension and shows all five seeds plus their mean. The individual source PDFs are also included under `figures/real_per_d/`.

which factors U 's spectrum into the spectra of $W^\top W$ and M_t , the latter inheriting the block-diagonal geometry of Σ_{data} .

Block decomposition of M_t . Direct computation gives

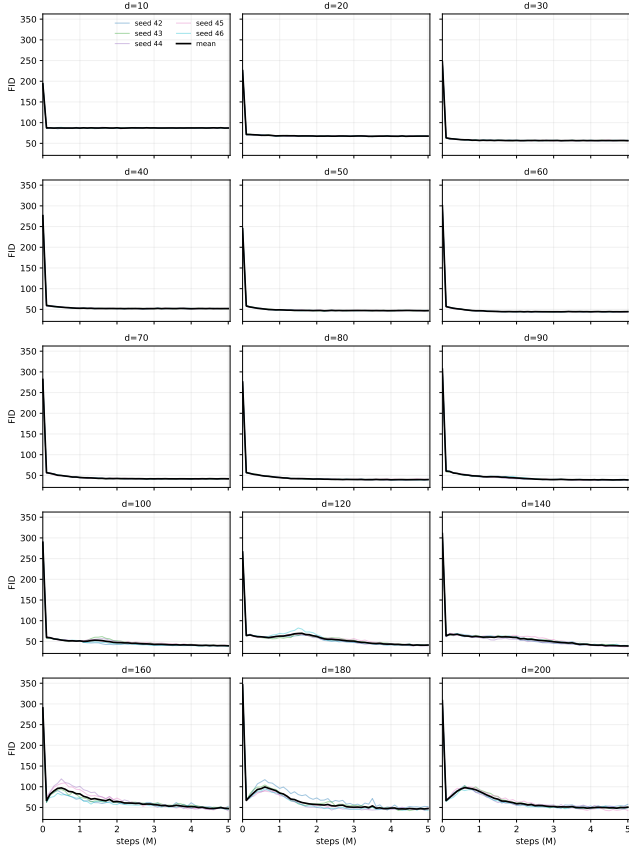
$$M_t = e^{-2t}(\hat{C} + \hat{S}) + \Delta_t \mathbb{I}_{d_{\text{latent}}}, \quad (7)$$

where $\hat{C} = \frac{1}{n} \sum_{\mu} m_{c(\mu)} m_{c(\mu)}^\top$ is the empirical centroid Gram matrix and $\hat{S} = \frac{1}{n} \sum_{\mu} \xi^{\mu} (\xi^{\mu})^\top$ is the empirical within-cluster covariance. In population, $\mathbb{E}\hat{C} = (1/k) \sum_c m_c m_c^\top$ has rank $d_{\text{intrinsic}}$ with eigenvalues $\Theta(s^2/d_{\text{intrinsic}})$ in the signal subspace; $\mathbb{E}\hat{S} = \Sigma_{\text{data}}$ has eigenvalues σ_{sig}^2 on the signal block and $\sigma_{\perp}^2$ on the null block. The sample fluctuations $\hat{S} - \Sigma_{\text{data}}$ are rank- n with spectrum given by a Marchenko-Pastur-type law (Karoui, 2010; Couillet & Debbah, 2011) of typical scale $\Sigma_{\text{data}}/\sqrt{n}$ in operator norm.

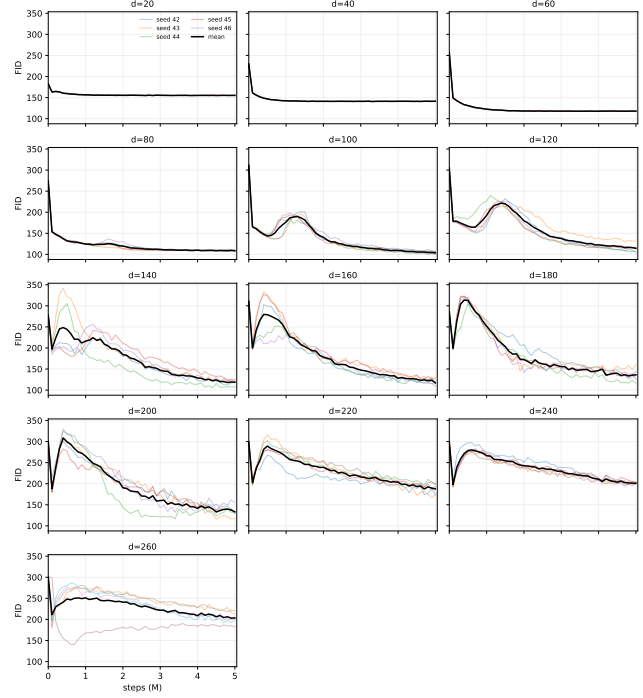
Four bulks from rank addition. M_t thus splits into three rank classes:

- (a) *Population signal* ($d_{\text{intrinsic}}$ modes): eigenvalues $e^{-2t} [s^2/d_{\text{intrinsic}} + \sigma_{\text{sig}}^2] + \Delta_t$, dominated by the cluster scale.
- (b) *Population null* ($d_{\text{latent}} - d_{\text{intrinsic}}$ modes): eigenvalues $e^{-2t} \sigma_{\perp}^2 + \Delta_t$, set by either the null-space data variance or the diffusion noise floor, whichever is larger.
- (c) *Sample fluctuation* (n modes): eigenvalues of $e^{-2t}(\hat{S} - \Sigma_{\text{data}})$ projected onto the signal subspace, scaling as $\Theta(e^{-2t} \sigma_{\text{sig}}^2/\sqrt{n})$, plus a smaller contribution in the null subspace.

Lifting through W via Eq. (6) and noting that $W^\top W/d_{\text{latent}}$ has bulk spectrum on $[(\sqrt{p/d_{\text{latent}}} - 1)^2, (\sqrt{p/d_{\text{latent}}} + 1)^2]$ for $p \gg d_{\text{latent}}$ (Pennington & Worah, 2017), U^{lin} inherits the three rank classes from M_t plus a fourth, trivial *rank-null*

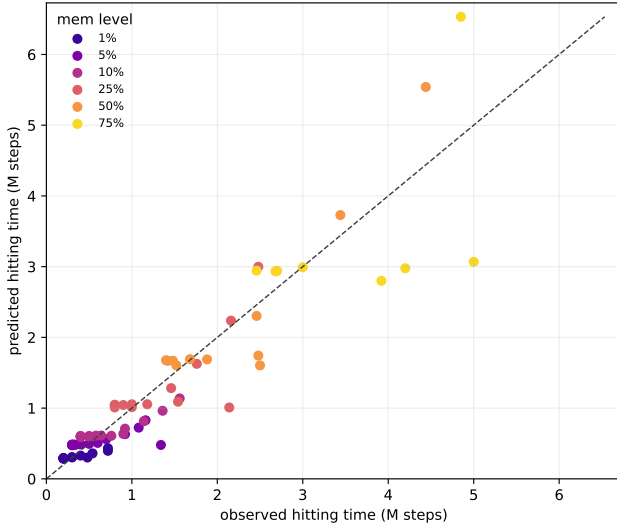


(a) CelebA FID.

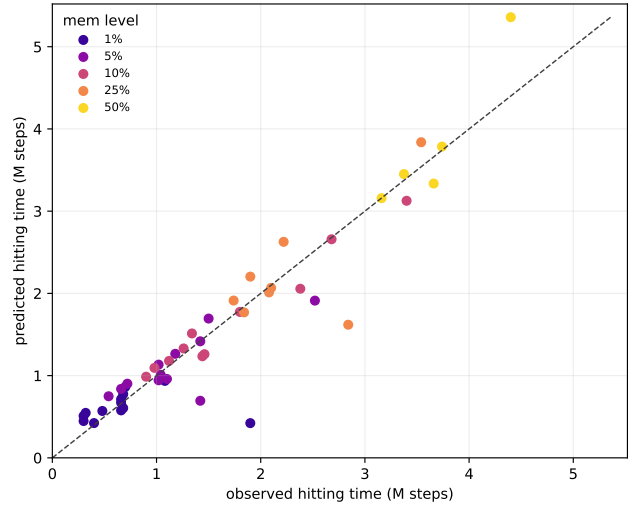


(b) CIFAR-10 FID.

Figure 27. **Per- d_{latent} FID trajectories.** These are the per-dimension panels underlying the aggregate FID curves in the body.



(a) CelebA.



(b) CIFAR-10.

Figure 28. **Predicted versus observed hitting times.** Each point is an iso-memorization threshold for a latent dimension. The predictor is calibrated globally across dimensions within a dataset.

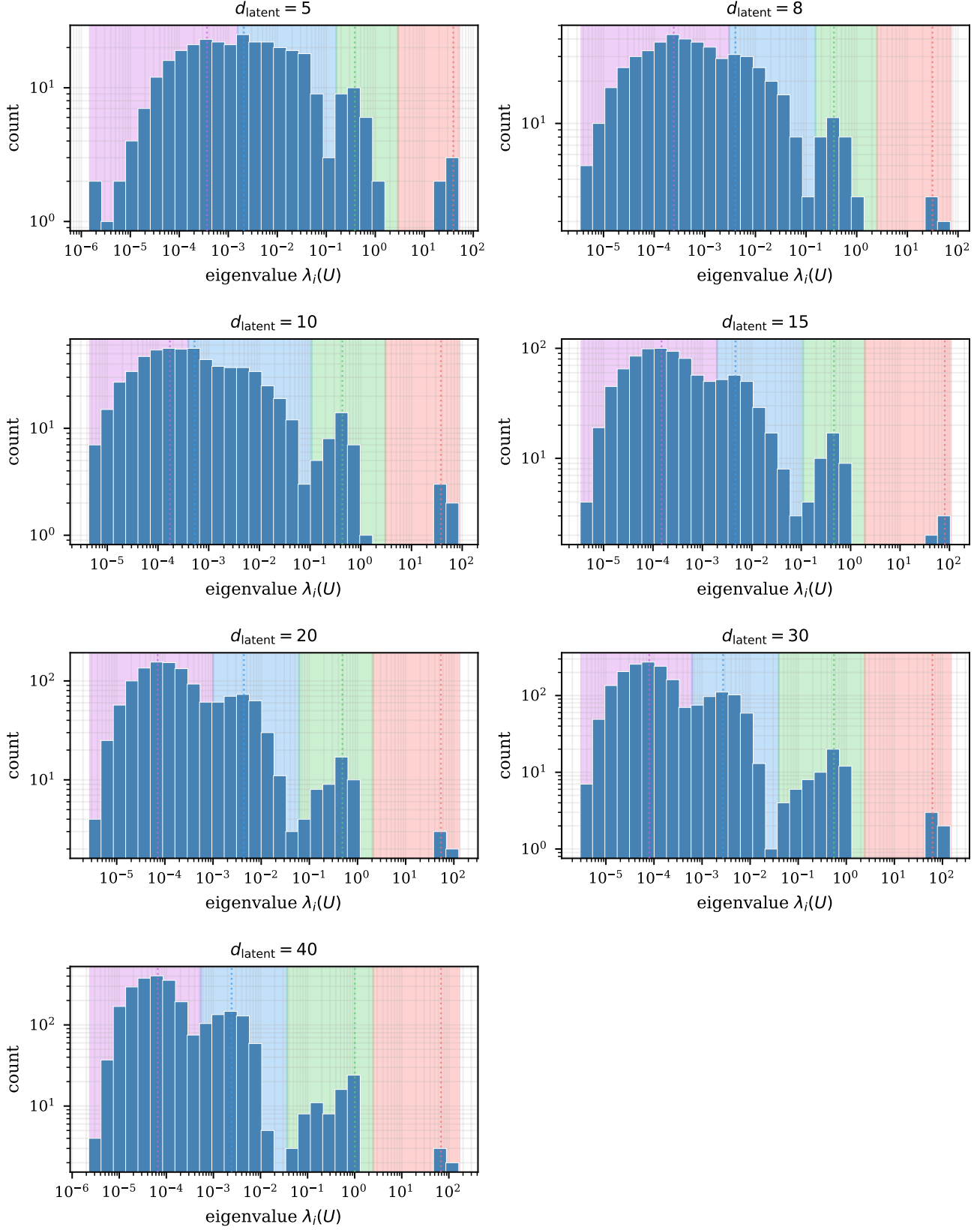


Figure 30. **Empirical bulk detection across the d_{latent} sweep at $\sigma_{\perp} = 0.01$.** Each colored region is one detected peak in the eigenvalue density; four bulks are detected in every panel. As $d_{\text{latent}}/d_{\text{intrinsic}}$ grows, the bulks pull apart: at $d_{\text{latent}} = d_{\text{intrinsic}}$ the four peaks sit close together with shallow valleys, and by $d_{\text{latent}} = 40$ ($d_{\text{latent}}/d_{\text{intrinsic}} = 8$) they have separated into well-isolated peaks decades apart in eigenvalue. Per-panel bulk sizes are tabulated in Table 3; detection algorithm in Appendix A.

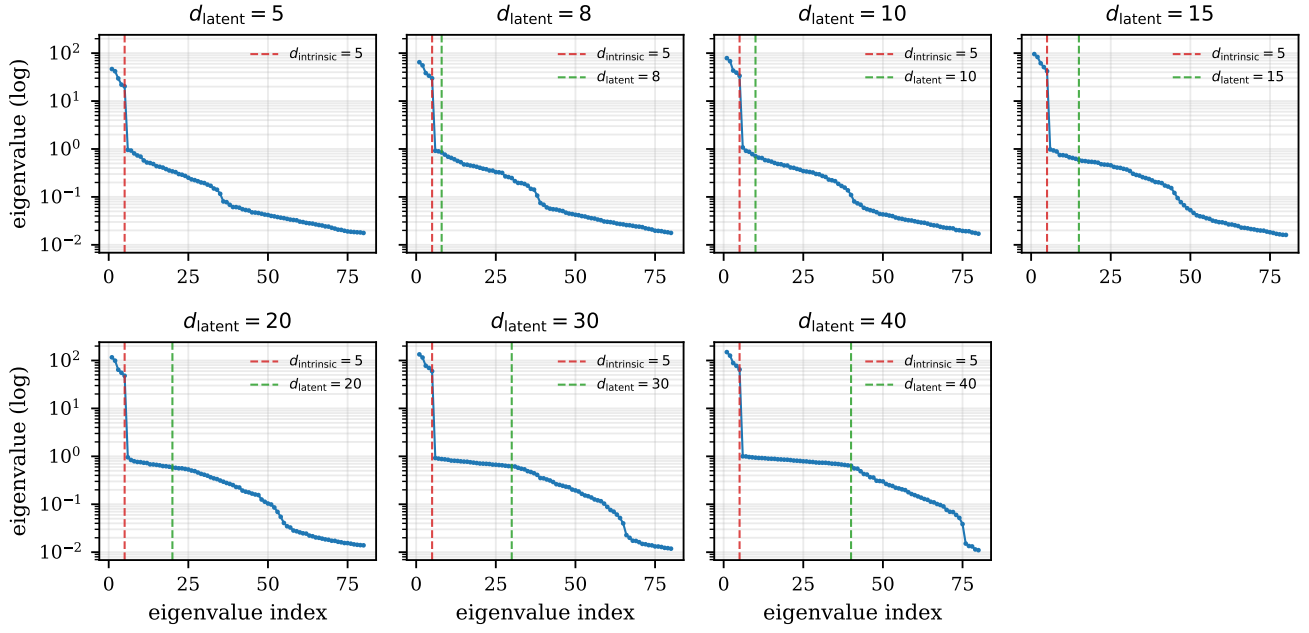


Figure 31. **Cliffs view**, $\sigma_{\perp} = 0.01$. Sorted log-spectrum across the d_{latent} sweep. Red and green dashed lines mark indices $d_{\text{intrinsic}}$ and d_{latent} . The BULK_DATA_NOISE region widens between them as d_{latent} grows.

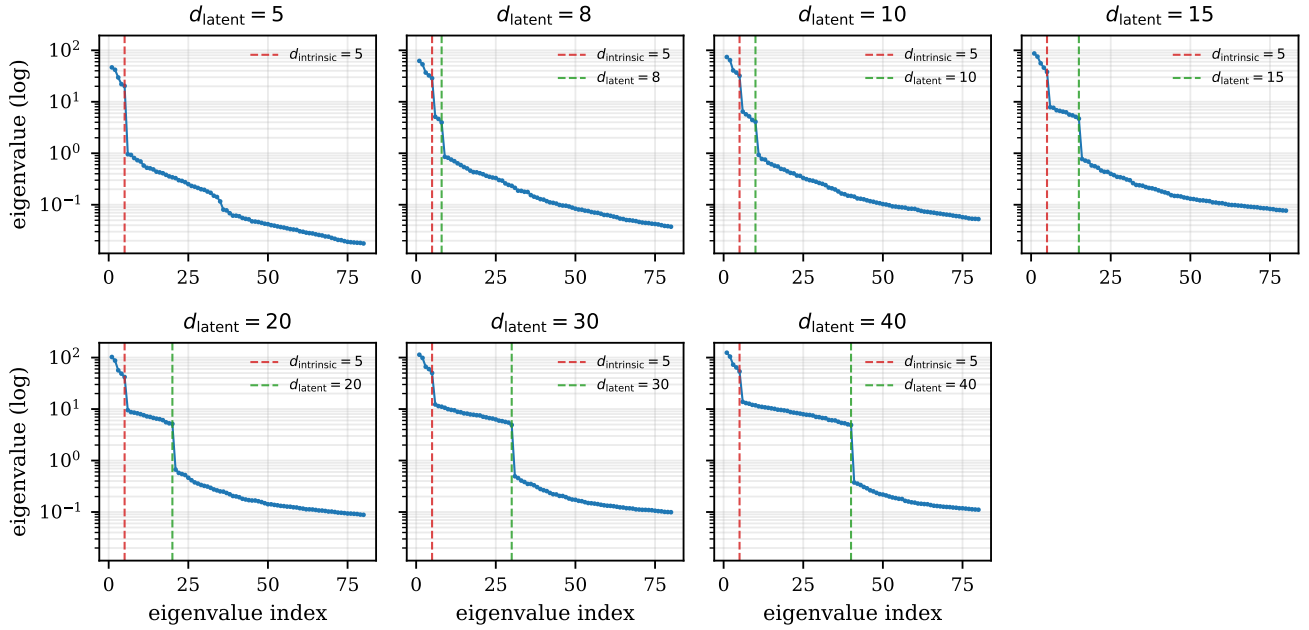


Figure 32. **Cliffs view**, $\sigma_{\perp} = 0.5$. Same sweep as Figure 31 at the higher noise scale: the signal/noise-dim cliff at index $d_{\text{intrinsic}}$ is sharp, while the data/sample cliff at index d_{latent} softens as the gap $\sigma_{\text{sig}}^2/\sigma_{\perp}^2$ shrinks.

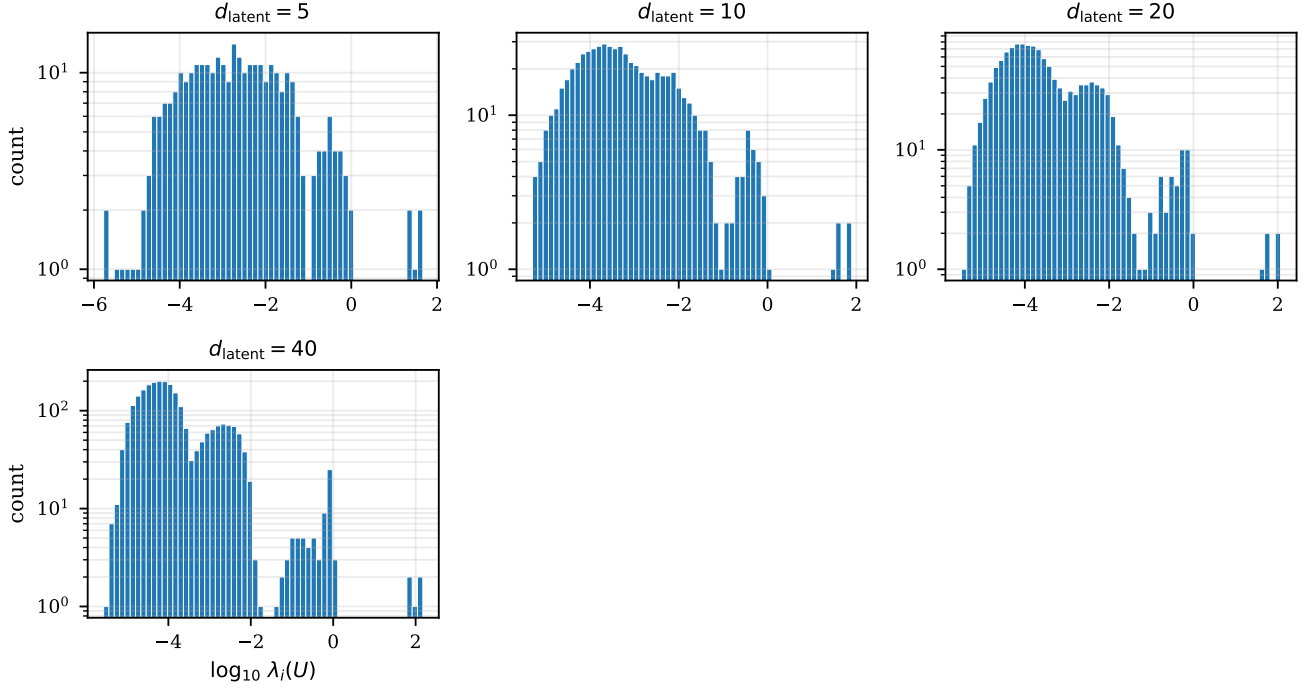


Figure 33. **Density view**, $\sigma_{\perp} = 0.01$. Counterpart to the body's Figure 2. The larger $\sigma_{\text{sig}}^2/\sigma_{\perp}^2 \approx 10^4$ pushes the signal bulk decays to the right of the noise-dim bulk.

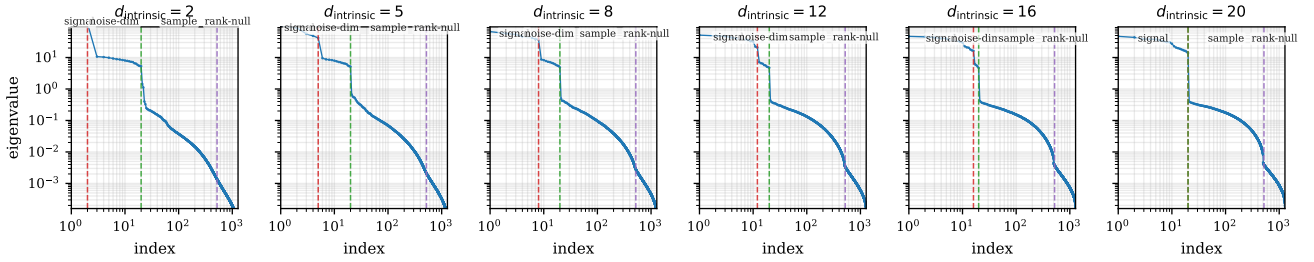


Figure 34. **Varying $d_{\text{intrinsic}}$** at $d_{\text{latent}} = 20$, $\sigma_{\perp} = 0.5$. As $d_{\text{intrinsic}} \rightarrow d_{\text{latent}}$ the noise-dim bulk shrinks toward zero width and we recover Bonnaire's two-bulk picture.